English Interjections Across Grammar, Meaning, and Interaction: Field-Relative Categories and Path Dependence

Brett Reynolds *
Humber Polytechnic & University of Toronto

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Abstract

English linguists use INTERJECTION for three overlapping but non-identical classifications: a grammar-facing lexical category, a semantics-facing category of expressive or use-conditional signs, and an interaction-facing category of conventionalized sequential resources. I argue that none has priority over the others. Each cuts through a mixed network of prosodic, morphological, syntactic, semantic, interactional, and historical relations. Each preserves different distinctions and generates a different family of projective hypotheses. Their common bearers are conventionalized sign-types: recurrent pairings of form with grammatical realization, meaning, and interactional affordance. Canonical cases belong to all three categories, but their overlap isn't a fourth category and supplies no independent projective force.

The empirical argument concerns path dependence: recruitment histories remain visible in source-specific residues. Invocative, verbal, and onomatopoeic routes, documented to different degrees, converge on recurrent combinations of properties. The history of *gee*, residual complement structure, euphemistic deformation, and semantic persistence in selected verbal derivatives support parts of this account. A preregistered study of *fie* fails to support ordered erosion, while GloWbE and COHA probes establish more limited patterns of country-level frequency variation and inventory turnover. These results support path dependence and motivate parts of a causal ordering. They also yield explicit projectibility hypotheses: predictions from observed properties to unobserved meaning and interactional treatment. Their held-out predictive performance remains to be tested directly.

Keywords: interjections; field-relative categories; projectibility; causal networks; supplements; path dependence

AI USE. The large language models Claude Opus 4.8, Claude Opus 5, and GPT-5.6 Sol served as drafting and editing aids throughout the preparation of this paper. I am responsible for all theoretical claims, arguments, errors, and interpretive choices.

*Contact: brett.reynolds@humber.ca

1 Introduction

English interjections are small forms with large consequences. A speaker's *oh*, *wow*, *hey*, *damn*, or *mm-hm* can package a response, display stance, open or sustain a sequence, initiate repair, or invite recipient uptake through a response or other displayed treatment. Grammars have never been settled about what to do with them. This paper argues that part of the difficulty is hidden by the singular question "Is interjection a category?" Grammar, semantics, and interactional pragmatics use the same term for three overlapping classifications with different extensions and different inferential purposes.

In 1586, William Bullokar wrote the earliest grammar of English. It included interjections, which he defined as words that betoken "a sudden passion of the mind" (Bullokar, 1586/1906, p. 373).

Four centuries later the standing is less secure. Jespersen (1924, p. 90) treated interjection as a manner in which words of other categories may be used, which is to deny it categoryhood. Quirk et al. (1985, p. 67) kept the category but called it "marginal and anomalous". Huddleston and Pullum (2002, p. 22) listed interjection as one of nine lexical categories in *The Cambridge Grammar of the English Language* (*CGEL*) (*ah*, *damn*, *gosh*, *hey*, *oh*, *ooh*, *ouch*, *whoa*, *wow*), while the companion textbook set interjections aside from its eight-category inventory (Huddleston & Pullum, 2005, p. 16). Its second edition returns interjection to the inventory, but concedes that "grammarians typically say very little about interjections, and we don't plan to be an exception" (Huddleston et al., 2022, p. 24). Readmitted to the list, and still largely undescribed.

Libert (2019) objects that calling interjections neglected "is an overstatement": work on their phonetics, syntax, semantics, and pragmatics goes back a long way. What's stayed unsettled is standing rather than volume of attention: whether grammar has a lexical category of interjections, whether the same classification serves semantics and interaction, and what any of those memberships would license.

Cram (2008, p. 63) makes the historiographic point more sharply. Surveying the tradition from Isidore of Seville onward, he finds "no firmer consensus about this category than at any earlier period", and notes that interjections "function as a touchstone or test case for various boundary controversies", naming the syntax–pragmatics and semantics–pragmatics boundaries. The difference is in what the case is asked to reveal. Cram asks what interjections show about our theories. I ask what holds these classifications together and what makes them diverge.

The empirical picture hasn't simplified. Evidence bearing on the traditional domain is spread across phonology, morphology, syntax, semantics, and pragmatics. No single property is necessary and sufficient, and the boundaries with nouns, verbs, adverbs, fillers, and routine formulae are all disputed. Cross-linguistically, "interjection" is at best a comparative concept (Haspelmath, 2010). But it doesn't follow that only a morphosyntactic classification can be a category. A semantic or interactional classification has equal standing if it supports reliable distinctions and inferences for its inquiry.

The relevant standard is PROJECTIBILITY: the inferential payoff by which observing some properties of a new case warrants expectations about others (Goodman, 1955). This paper specifies such expectations for each interjection category. It doesn't estimate their held-out predictive performance. The present evidence instead bears on the architecture that generates them: recurrent clustering, documented and reconstructive recruitment pathways, patterned residues, and cases where the three classifications come apart.

A form's PROFILE is the particular combination of cluster properties it shows in a given grammat-

ical environment: whether it's prosodically set off, whether it inflects, whether it refers, what stance or response it carries. The word is doing bookkeeping here and has nothing to do with the figure-and-base sense familiar from cognitive grammar. To say that a property or a category LICENSES an inference is to say the inference is a good default, defeasible by the particular case, not that it follows necessarily.

The object of analysis is a conventionalized linguistic SIGN-TYPE: an expression paired with a conventional profile of grammatical realization, meaning, and interactional affordance. The three classifications apply to that common bearer. Thus verbal *damn* in *I damn you*, standalone *damn!*, complement-taking *damn these mosquitoes!*, and prenominal *damn* in *the damn car* are distinguishable sign-types even though they belong to one historical word family.

By PATH DEPENDENCE, I mean that where a form came from keeps showing in how it behaves at the category's edges, long after it has arrived. *Gee* is documented as developing through a religious invocation and takes no complement. Complement-taking *damn* is compatible with a verbal optative source and retains NP complementation (*damn these mosquitoes!*). The second history remains partly reconstructive, but the contrast illustrates the test: if convergence erased history, present-day residues shouldn't align with independently supported sources.

Throughout, I use PROPERTY CLUSTER descriptively for an empirical pattern of co-occurring features. A cluster isn't automatically a category, and recurrent co-occurrence isn't automatically causal order. Those further claims require their own evidence.

This inferential framing motivates a causal-network analysis. Khalidi (2013, 2018) argues that natural kinds are nodes in causal structures rather than sets defined by essences. On this account, a category is theoretically useful when its place in a causal network supports reliable induction. Applied to interjections, the proposal is that diagnostics including prosodic isolation, syntactic independence, non-referentiality, non-inflection, and interactional force are connected through multiple local links and historical pathways. The strongest present evidence concerns recruitment and residue. Other links remain ordering hypotheses or stabilizer candidates.

Four things have to be kept apart, because they fail separately (Reynolds, 2026c). Each has a label I'll use throughout.

- The PROJECTIVE DECLARATION states the claim: what's observed, what's predicted from it, and over which cases. Here a source is a specified field-relative classification supported by a restricted set of observed diagnostics. A target is a genuinely unobserved property or environment, the bearer is a conventionalized sign-type, and the population is present-day English. A target may belong to a category's general individuation profile, but it can't also be part of the case-level evidence used to make that declaration.
- WARRANT is the separate question of why that inference is licensed and how strongly: what evidence supports it, and what would defeat it. Table 2 records this link by link.
- The profile just defined is the worldly relation the declaration posits, and STABILITY is the further claim that it recurs across forms and over time. NETWORK ORDER is a claim of a different kind: that the properties are linked, so that one makes another more available. *Stability and network order are independent.* A combination can recur without revealing any direction among its parts, and a directed dependence can hold without recurring.
- GROUNDS explain why the profile obtains or persists (Reynolds, 2026b). STABILIZERS are grounds that hold the combination together as forms enter and leave. CORRECTIVE CONTROL

is the strongest, a stabilizer that restores the combination when something disturbs it. Only that would license calling the cluster HOMEOSTATIC, Boyd's term (Boyd, 1999), and the evidence here doesn't establish it.

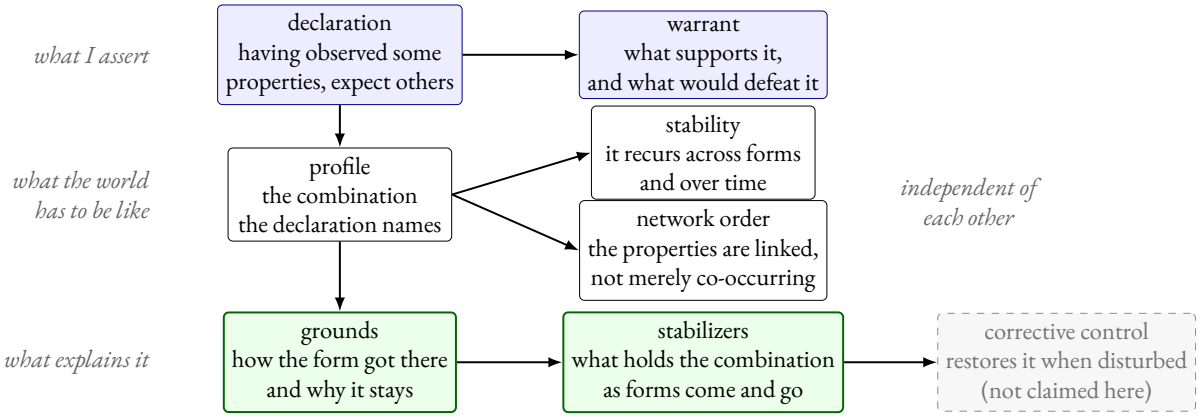


Figure 1: The four objects a category claim has to keep apart. The declaration is the commitment; the warrant is the case for it. The profile is what the world has to supply, and stability and network order are further claims of different kinds, neither following from the other. Grounds explain the profile, stabilizers hold it together, and corrective control is the strongest ground of all. Only corrective control would license calling the cluster HOMEOSTATIC, and it isn't claimed here.

One form shows what that comes to. Take *gee*, as in (1).

(1) *Gee, that's a lot of money.*

Observing that *gee* is set off from what follows and never inflects, with no **geed* and no **gees*, supplies the source side of a possible declaration. The proposed targets are that the sign-type will work as a supplement in unobserved environments, contribute use-conditionally rather than descriptively, and be available at the front of a turn. Failure on those targets would defeat the corresponding projections. Complement-taking or inflection would instead undermine the source classification and so shouldn't be counted again as failed target behaviour.

Diachrony matters because pragmatic profiles have histories. Forms such as *gee*, *damn*, or *bruh* change category position as historically contentful or addressive expressions come to manage stance, sequence, and interactional alignment in recurrent contexts (cf. Jucker & Taavitsainen, 2013). Change over time helps explain why recurrent profiles and source-specific residues can coexist while individual forms enter and leave an inventory.

The network classifies sign-types, not word families as wholes. I write the three categories as INTERJECTION$_{\text{SYN}}$, INTERJECTION$_{\text{SEM}}$, and INTERJECTION$_{\text{PRAG}}$. Bare "interjection" remains a conventional cover term for the traditional domain and for canonical cases that satisfy all three classifications. Their overlap does no independent classificatory or projective work: joint membership simply makes the three component families of inference available. Section 3.4 defines the categories and explains why their extensions overlap but come apart.

Section 2 documents the cluster. Section 3 introduces the causal-network framework and the three field-relative categories. The causal sections grade the proposed links by evidential status and

develop the strongest present payoff, path-dependent residue. The projectibility section then states the forward inferences the architecture generates and the evidence still needed to establish their held-out success.

2 The property cluster

English interjections exhibit a cluster of co-occurring properties. This section takes phonology, morphology, syntax, semantics, and pragmatics in turn, drawing primarily on the empirical base documented in Ameka (1992), Quirk et al. (1985), and Huddleston and Pullum (2002). It answers two questions: what properties cluster, and what local diagnostic inferences each property suggests. Six properties emerge: prosodic isolation, non-inflection, syntactic non-integration, non-referentiality, emotive or stance force, and sequential use, the last being the capacity to constitute a conversational move on its own. The three categories cut across those six rather than regrouping them into domain-pure bundles, because each inquiry preserves different contrasts and proposes different targets. The subscripts are used from here on, with assignment criteria given in Section 2.6 where the diagnostic matrix needs them, and the argument for treating the categories as three deferred to Section 3.4. Section 5 asks the stronger question: when does a partial observation license inference across linguistic domains? The causal question, why these properties travel together, is deferred to Section 4.

2.1 Phonology

Interjections are prosodically isolated, and Ameka (1992, p. 108) states it without hedging: they're "always separated by a pause from the other utterances with which they may co-occur" and "always constitute an intonation unit by themselves". I treat the property as strongly characteristic rather than exceptionless, because the diagnostic matrix below records partial cases (*um*). That gap between a categorical statement and its boundary cases is the pattern Section 2.6 takes up. In writing, this isolation surfaces as punctuation (commas, dashes, exclamation marks), setting the interjection off from the clause it accompanies (Huddleston & Pullum, 2002, p. 1350).

The corpus evidence below is written, so prosodic isolation is tracked through proxies: punctuation, standalone position, and absence of integration into the host clause. These proxies are useful for historical and large-corpus work, but they're proxies rather than direct acoustic evidence. They aren't arbitrary ones. On Jackendoff (2025, p. 41)'s account a parenthetical is realized as an intonation phrase, so it ends in a boundary tone made of "some combination of pause, lengthening, pitch discontinuity", and the flanking commas are what record those boundaries. Punctuation tracks prosodic isolation because it notates the same breaks.

Interjections may also contain phonemes outside the speaker's dialect. The interjection *uh-oh* ([əʔo]) is one of the few cases where a glottal stop appears in dialects that otherwise lack it. Other atypical sounds include the alveolar clicks in *tut-tut* ([ǃǃ]), the voiceless bilabial fricative in *whew* ([ɸɪu]), and (for dialects that no longer use it) the voiceless velar fricative in *ugh* ([əx]) (Quirk et al., 1985, p. 853). Spelling pronunciations often arise for these forms (*tut-tut* said [tət tət], *ugh* said [əg]), suggesting that the atypical phonemes are marked even for native speakers.

Prosodic isolation is a strong cue: if a novel form occupies its own intonation unit and resists integration into the surrounding prosodic contour, the network account proposes syntactic non-integration and non-referentiality as targets. Their held-out relation to that cue remains to be tested. Prosodic isolation isn't unique to interjections, since vocatives, appositives, and other supplements

share it (cf. Wilkins, 1992). What distinguishes interjections is the convergence of prosodic isolation with the properties below.

2.2 Morphology

English interjections characteristically don't inflect or derive. They resist the morphological operations that characterize other open-class categories: no plural (**ouches*), no tense (**ouched*), no comparative (**oucher*), no nominalization (**ouchness*) (Ameka, 1992, p. 106). Since Libert (2019) notes that "few, if any, papers focus on the morphology of interjections", the apparent absence of morphological productivity may partly reflect a gap in the research rather than a firm empirical generalization.

Some interjections retain remnants of earlier categorial membership. *Heavens* and *bollocks* preserve the nominal plural suffix. None of these remnants are productive. The plural in *heavens* doesn't contrast with a singular **heaven!* (as an interjection), and speakers don't extend the pattern to new forms.

Non-inflection is associated with restrictions on ordinary argumenthood in the recurrent profile. It doesn't entail them: adverbs also don't inflect, and invariance alone doesn't prevent an expression from filling an argument slot (Ameka, 1992, p. 106). The stronger diagnostic is syntactic: adverbs characteristically form constituents with other words, while canonical interjections occur as standalone supplements and resist integration with neighbouring material (Meinard, 2015, p. 152).

2.3 Syntax

Interjections characteristically function as supplements, "elements which occupy a position in linear sequence without being integrated into the syntactic structure of the sentence" (Huddleston & Pullum, 2002, p. 1350). In *Damn, we're going to be late!*, the interjection isn't a complement or modifier of anything in the clause. It's appended to the beginning as a parenthetical.

Supplement function isn't unique to interjections. Relative clauses, noun phrases, adjective phrases, and adverb phrases can all function as supplements (Huddleston & Pullum, 2002, pp. 1356–1360):

(2) a. *Ah, so you were there after all!* (interjection, p. 1360)
b. *We called in to see Sue's parents, which made us rather late.* (relative clause, p. 1356)
c. *A university lecturer, Dr Brown, was arrested for the crime.* (noun phrase, p. 1357)

What distinguishes interjection supplements is their combination with the other cluster properties: non-referentiality, non-inflection, prosodic isolation, and emotive force. The function is shared; the category differs.

A handful of interjections exceptionally take complements. Huddleston and Pullum (2002, 1361fn) analyse *damn* in *damn these mosquitoes!* as the head of an interjection phrase (IntjP) taking a noun phrase (NP) complement, one of the very few cases where an interjection heads a phrase larger than a single word. But even here the syntax is constrained: no subject is overt, and there's no tense inflection. The construction is compatible with a residue of verbal source syntax, a reconstructive possibility that matters for the recruitment pathways discussed in Section 4. A second kind of residue is internal rather than external. *Goddammit* preserves a verb-object-pronoun structure inside what's now a single word, so the syntax survives as fossilized material rather than as anything the form can still do. Cuenca (2002, p. 40) reads it the same way from the grammaticalization side, listing *Goddammit* among the cases where change in function and meaning is "accompanied by

phonetic weakness and word integration". (The internal structure of lexicalized pairs like *oh boy* and *holy cow* is less clear and won't be resolved here.)

Supplement function predicts prosodic isolation and resistance to syntactic integration:

(3) a. * *I think that damn we're going to be late.*
b. *I think that, damn, we're going to be late.*
c. * *Damn and quickly, we left.*

The contrast in (3) is prosodic rather than syntactic: the first two differ only in whether *damn* is set off. Jackendoff (2025, pp. 41–42) generalizes the point beyond interjections as the IP-Break Hypothesis: placement tracks intonational-phrase breaks rather than syntactic constituency, so a host admits a parenthetical where it independently permits an IP break, provided the parenthetical is itself licensed in that position (Jackendoff, 2025, p. 43). He gives pairs of hosts that are "syntactically identical" but differ in where a parenthetical may go, since "the difference in the parenthetical's position is a consequence of the difference in position of the IP breaks in the host" (Jackendoff, 2025, p. 42).

Supplement function also predicts a semantic consequence: if the form is syntactically non-integrated, its contribution is independent of the proposition expressed by the host clause. Remove the interjection and the truth conditions don't change, the hallmark of non-truth-conditional meaning (cf. Potts, 2007, p. 166).

2.4 Semantics

English interjections characteristically don't refer. Here, NON-REFERENTIALITY means absence of ordinary entity/event denotation and argumental construal while allowing semantic content. Nouns and verbs typically pick out entities or events in the world. Interjections express internal states of their users (Leech, 2006): anger (*damn*), disgust (*eww*, *yuck*), surprise (*wow*), regret (*alas*), pain (*ouch*), realization (*eureka*). Some interjections are more propositional than emotive: *yes* and *no* affirm or deny propositions, and *oh* can signal recognition of new information rather than expressing feeling.

Wierzbicka (1992) argues that interjections have genuine semantic content, decomposable via Natural Semantic Metalanguage. The forms may have content, and *yes* and *no* may operate on propositions, while still lacking ordinary entity/event denotation.

Wilkins (1992, p. 119) presses harder, and his title states the objection: interjections are deictics. On his account they "convey complete propositions", the "referential arguments in the propositions conveyed by interjections are provided by context", and so interjections are "shifters (indexicals) by virtue of being built semantically out of basic deictic elements" such as *I*, *you*, *now*, and *here*. If that's right, calling interjections non-referential looks like a straightforward mistake.

The two claims are about different things, and Wilkins's own wording shows it: the referential arguments are supplied by context, not by the form. Non-referentiality, as I use it, is a claim about what a form contributes on its own, so it leaves the contextual filling untouched. *Ouch!* picks out no thing or event, and resists being made definite, quantified over, or used as an argument, whatever indexical structure its explication turns out to contain. If anything Wilkins asks more of the semantics than I do, since he has to say what it supplies for the filling to work at all, and the distributional facts I rely on hold on his account as readily as on mine. What his account does show is that "non-referential" is a poor name for the property, kept here only because it's the established one.

Not everyone reads Wilkins that way. On the conceptualist analyses of Wierzbicka (1992), Ameka (1992) and Wilkins, interjections encode substantial conceptual structure. Wharton (2000,

p. 211) argues from relevance theory that they encode considerably less, being "partly natural and partly coded" and falling "at various points along a continuum between showing and saying". That dispute doesn't need settling here. Neither Wilkins's deictic slots nor Wharton's activated attitudinal concepts give *ouch!* a thing or event to pick out, so the distributional property holds on either analysis. But Wharton's continuum is worth noting for a second reason. What varies along it is how much of an interjection's import is coded rather than shown, which is a gradient within the category and not a blurred edge to it (Section 2.6).

Cruz (2009, p. 241) carries the dispute further. He argues that the heterogeneity of interjections, "with items incorporated from other lexical categories", together with the openness of the class, "could indicate the existence of a continuum of more and less conceptual items", and that the conceptual ones encode "not full concepts, but some schematic material requiring subsequent pragmatic adjustments". What matters here isn't the continuum, which Wharton already supplies, but its cause. Cruz derives gradedness from recruitment out of other categories and from open-class coinage, which are the mechanisms Section 4 treats as the cluster's sources.

Wierzbicka's explications characterize the use-conditional content that INTERJECTION$_{SEM}$ (Section 3.4) tracks. Non-referentiality is a cluster property rather than a definition.

The contrast with nouns is sharpest where both share a lexical source. *Jesus* as a noun refers to a person; *Jesus!* as an interjection doesn't. In the interjection use, as Gehweiler (2008, p. 88) puts it, *Jesus* "has lost its propositional content, i.e. it is no longer used to refer and hence no longer influences truth conditions". This semantic shift is part of the recruitment pathway from noun to interjection (Section 4).

The projectibility here is mostly negative. A form that doesn't refer resists being made definite, quantified over, or used as an argument: **the damn!*, **every ouch*, **Damn surprised me*, and **There's damn*. These restrictions are non-trivial because they distinguish interjections from nouns that have undergone only partial bleaching. Section 4.5 treats the same relation from the causal side: non-referentiality helps keep the form out of argument structure, which in turn supports prosodic isolation.

2.5 Pragmatics

Interjections typically carry exclamatory or directive force: *ouch!* expresses pain, *shh!* demands silence (Quirk et al., 1985, p. 88). But the pragmatic range extends beyond exclamation. How far beyond is worth a number. Elffers (2008, p. 18) reports that in Hofstede's analysis of a corpus of spoken Dutch, only 7.04 % of interjections "turned out to fulfil an expressive illocutionary function", against 52 % classed as expositive, which he glosses as directive within the conversation. The corpus is Dutch rather than English, so the figure indicates rather than establishes. It points the same way as Dingemanse's interactional focus. The emotive characterization grammars lead with covers the minority case. Interjections greet (*hey*, and, at the boundary with routine formulae, *hello*; see Section 6.2), mark agreement (*yes*, *amen*), indicate understanding (*oh*, *uh-huh*), and signal backchanneling, the listener's ongoing attention to a speaker's turn (Dingemanse, 2020; Stivers, 2019).

The most common interjections tagged in the Corpus of Global Web-Based English (GloWbE; Davies & Fuchs, 2015) illustrate this pragmatic breadth: *yes*, *no*, *oh*, *yeah*, *hi*, *hey*, *wow*, *hello*, *ah*, *ha*. The list is dominated by interactional items, forms whose primary function is to manage the conversation rather than to contribute propositional content. Poggi (2009, p. 171) presses the other way. An interjection, on her analysis, is a holophrastic signal whose equivalent isn't a word but "a commu-

nicative act including the meaning of both a performative and a propositional content", so that *ouch!* paraphrases as "I am feeling pain". The grain distinction from Section 2.4 applies again. Poggi's propositional content belongs to the act the interjection performs, not to any clause it helps compose. Her contrast between holophrastic and articulated signals says why. Nouns and verbs characteristically convey part of a sentence, an interjection a whole one. Prenominal *damn* in *the damn car* is an interjection that doesn't. That explains the independence noted in Section 2.3: a form contributes nothing to a host clause's truth conditions if it isn't part of a clause at all. Holophrasis accounts for the property rather than competing with it. Dingemanse (2024) estimates that roughly one in seven conversational turns is an interjection, and interactional tools dominate the most frequent cases: continuers like *mm-hm* that scaffold extended turns, repair initiators like *huh?* that calibrate mutual understanding, and change-of-state tokens like *oh* that display evolving knowledge[1] (cf. Norrick, 2009).

The boundary with routine formulae is instructive. Coulmas (1981, pp. 2–3) defines routine formulae as "highly conventionalised prepatterned expressions whose occurrence is tied to more or less standard communication situations": *bye*, *hello*, *sorry*, *thank you*. Ameka (1992, pp. 109–110) argues these differ from interjections in three ways: they include addressees, they're predictable responses to social conventions, and they're always speech acts rather than mere reflections of the speaker's mental state. Wilkins (1992, p. 142) counters that these differences make routine formulae "a distinct pragmatic and semantic subtype of interjections", not a separate category.

The disagreement is field-relative in a causal-network sense. For a grammar, routine formulae pattern near interjections because they share prosodic isolation and pragmatic function. For speech-act analysis or interactional pragmatics, they support different predictions: addressee orientation, occurrence in standard social sequences, and conventional felicity conditions. The same form can sit close to one node for one inquiry and farther away for another, depending on which causal relations and predictions the inquiry tracks.

Which projective declarations are available depends on what was observed. A semantics-facing classification of a novel stance form may motivate targets in unobserved grammatical or interactional environments. A routine-formula classification may motivate addressee orientation or placement in a conventional social sequence, provided those properties weren't used to assign the case. The overlap explains why routine formulae sit at the boundary; the divergence explains why the classifications needn't agree. Section 5 fixes the source–target discipline.

2.6 Co-occurrence, not necessity

These properties recur together in the descriptive literature, but not necessarily. The literature's own definitions hedge, on every criterion they state. Wilkins (1992, p. 124) defines an interjection as "a conventional lexical form which (commonly and) conventionally constitutes an utterance on its own, (typically) does not enter into construction with other word classes, is (usually) monomorphemic, and (generally) does not host inflectional or derivational morphemes". Four criteria, a parenthetical hedge on each, and Gehweiler (2008, p. 73) adopts the definition while calling those criteria the class's "obligatory properties". No single property is necessary and sufficient.

[1] Token does double duty in work of this kind, and I keep the senses apart. In change-of-state token and response token it's the interactional-linguistic sense: a minimal form deployed as a move in a sequence. Elsewhere I use it for the unit of analysis, a lexical item in a grammatical environment as against the lexeme that labels a family of historically related such items.

One dissent belongs here, because it's the strongest case against that last sentence. Libert (2012, p. 282) proposes exactly one criterion: interjections have a CONVERSATIONAL VALENCE of 1, meaning they can be used felicitously with no addressee present, where *goodbye* requires one and so has a valence of 2. If you stub your toe you can say *ouch!* alone. On this account interjections are "a homogeneous word class, clearly delimited by a sole criterion" (Libert, 2012, p. 294).

The criterion works. What it costs is visible in how Libert reaches it, by redefinition: interjections "as I have redefined them, have only one function (as 'expressions of feeling'); the function of 'marks of address' is undertaken by another class" (Libert, 2012, p. 294). The addressing uses are moved out of the category rather than shown to satisfy the criterion, and they take a good deal with them: *hey*, *shh*, and *psst*, and also the continuers, repair initiators, and response tokens of Section 2.5, all of which need someone to be there. A single property is necessary and sufficient for what's left.

I take that as support for the decomposition proposed below rather than as a counterexample to co-occurrence. Libert's line between expressions of feeling and marks of address falls close to the line this paper draws between INTERJECTION$_{\text{SEM}}$ and INTERJECTION$_{\text{PRAG}}$, and he credits Ameka (1992, p. 109) with the same intuition while objecting, fairly, that Ameka's gloss of conative interjections as "directed at" but not "addressed to" people is obscure. So the disagreement isn't about the facts. It's about what to do with them: Libert narrows the category until one criterion suffices, and I keep the wider class and let the properties co-occur without any one of them being definitional. That isn't a bare preference. Categories in this tradition are established by clusters of grammatical properties rather than by single criteria, since a definition has to identify "the grammatical properties that distinguish them from the concepts with which they contrast" (Huddleston & Pullum, 2002, p. 30). A single criterion would make interjection the only category in the inventory picked out that way, which is a reason to doubt the standard rather than the category.

Co-occurrence without necessity doesn't settle whether membership itself is graded. The analysis needs a more local discipline. Each diagnostic has to be assessed for a specified sign-type: complement-taking *damn* either takes an NP complement in the construction at issue or it doesn't, and *heavens* either carries the plural suffix or it doesn't. Table 1 records categorical analytical decisions, with parentheses marking evidential qualification or a live classification dispute. Nothing in the residue, turnover, or projectibility arguments depends on a further metaphysical claim that every category boundary is sharp.

The pattern calls for explanation. The causal-network proposal connects the properties: prosodic isolation may reinforce syntactic non-integration, non-referentiality reduces the motivation for argument structure, and pragmatic conventionalization may entrench interactional functions.

The coupling is visible when it breaks. *Damn* as an interjection (*Damn!*) has the full cluster: prosodic isolation, syntactic non-integration, non-referentiality, and emotive force. Prenominal *damn* in *the damn car* retains emotive force, and with it non-inflection and non-referentiality, but loses prosodic isolation and syntactic non-integration together, because it's embedded inside a noun phrase (cf. Potts, 2007). Those two properties co-vary across this contrast, as the proposed coupling link predicts (Section 4.5). A bare property list records the contrast; a successful causal account would explain it. Two cases alone don't establish the link.

Table 1 makes the cluster inspectable. Each row gives a conventionalized sign-type rather than a word family as a whole (see Section 1), and the columns show its profile against the six cluster properties and the three field-relative categories defined in Section 3.4. The table shows convergence and

decoupling. Standalone *damn!* and *oh* in a change-of-state response show convergence. In *the damn car*, prenominal *damn* has the semantics-facing category but lacks the grammar-facing one. *mm-hm* has the interaction-facing category, while its grammatical status is debatable. Boundary items (*bye*, *um*) share some but not all cluster properties, the pattern that Section 6.2 explains as causal-link overlap.

These six recur across the sections above, but they aren't a complete inventory: free coinage by onomatopoeia distinguishes the category without belonging to any single item, so it has no column here. The last three columns are separate classifications, not sums of the first six, so they need their own criteria. A sign-type is Intj_{syn} if it functions as a supplement, taking no complement and forming no constituent with neighbouring material; Intj_{sem} if its contribution is expressive and use-conditional, leaving the truth conditions of its host untouched; $\text{Intj}_{\text{prag}}$ if it's conventionally usable as a minimal interactional resource in a specified sequential environment. This last criterion assigns the category from observed source environments; projection to recipient treatment or a withheld environment is a separate declaration. The semantic criterion is about meaning type rather than position, which is why a construction can satisfy it while sitting inside a host clause. Section 3.4 argues for the three-way division; here the columns record it.

Table 1: Diagnostic matrix: constructions against cluster properties and field-relative categories. ✓ = property present; (✓) = partial or debated; – = absent. Intj_{syn}, Intj_{sem}, and $\text{Intj}_{\text{prag}}$ abbreviate grammar-facing, semantics-facing, and interaction-facing categories. Sources: properties from Ameka (1992), Quirk et al. (1985), Huddleston and Pullum (2002); $\text{INTERJECTION}_{\text{SEM}}$ from Potts (2007); $\text{INTERJECTION}_{\text{PRAG}}$ from Stivers (2019), Dingemanse (2020).

	Prosodic isol.	*Non-inflecting*	*Synt. non-int.*	*Non-referential*	*Emotive/stance*	*Sequential use*	Intj_{syn}	Intj_{sem}	$\text{Intj}_{\text{prag}}$
gee!	✓	✓	✓	✓	✓	✓	✓	✓	✓
damn!	✓	✓	✓	✓	✓	✓	✓	✓	✓
damn these …!	✓	(✓)	(✓)	✓	✓	–	(✓)	✓	–
the damn car	–	✓	–	✓	✓	–	–	✓	–
oh (change of state)	✓	✓	✓	✓	✓	✓	✓	✓	✓
mm-hm (continuer)	✓	✓	✓	✓	(✓)	✓	(✓)	–	✓
huh? (repair)	✓	✓	✓	✓	(✓)	✓	(✓)	–	✓
bruh! (stance)	✓	✓	✓	✓	✓	✓	✓	✓	✓
bye (closing)	✓	✓	✓	(✓)	–	✓	✓	–	(✓)
um (filler)	(✓)	✓	✓	✓	–	(✓)	(✓)	–	(✓)

The next two sections develop the theoretical framework (Section 3) and the causal structure itself (Section 4).

3 The causal-network framework

The property cluster documented above has the shape Section 2.6 described: recurrent co-occurrence in the descriptive literature, no necessary and sufficient conditions, disputed boundaries, and hedged universals. The tradition drew the natural conclusion, that a class with no definition behind it has

nothing to license, and grammars have said little for that reason (Huddleston & Pullum, 2005, p. 16). The inference doesn't hold automatically. A category may instead earn its place by supporting warranted projections. The question is whether a causal-network analysis can specify such projections for the three interjection categories without treating any of them as an essence.

3.1 The framework

The relations that unify a kind may be local mutual-reinforcement relations, one-directional pathways, or converging historical routes (Khalidi, 2013, 2018). Each of the three has work to do below: the coupling links of Section 4.5 are mutual reinforcement, the non-referentiality ratchet is one-directional, and the recruitment pathways of Section 4.1 are converging historical routes. What matters is whether a category marks a position where properties and processes are connected tightly enough to support induction from observed to unobserved properties. That's the proposal to be evaluated, not a consequence of calling the relations a network.

I take the structure from Khalidi and not the metaphysics. Nothing below turns on whether interjection is a kind in a philosopher's sense; what the category has to do is support non-trivial, field-specific projections for linguistic inquiry. The present evidence motivates several such declarations and bears directly on recruitment and residue; it doesn't yet estimate the declarations' held-out performance.

The four objects distinguished in Section 1 are easier to see running on a single form. Take *oof* in standalone use, as in (4).

(4) *Oof. That's going to hurt.*

- A declaration built from restricted source evidence says this: having observed that the sign-type stands prosodically separate and doesn't inflect, expect it to resist determiners, to lack ordinary entity or event denotation, and to contribute use-conditionally rather than descriptively, in present-day English and in this use rather than across the whole word family.
- The warrant is whatever supports that expectation, and here it's thin: analogy to *ouch* and *wow*, and the links set out below.
- The profile is the combination the declaration posits. Stability and network order are further and independent claims: that the combination recurs across forms and over time, and that the properties are linked rather than merely co-occurring. Neither follows from one form.
- Grounds explain how the form got there and why it stays. *Oxford English Dictionary* (2026b) derives *oof* as an imitative formation, which puts its entry on the onomatopoeic route of Section 4.1 rather than on either bleaching path. A stabilizer would be whatever keeps the combination together as forms enter and leave, and corrective control would be a process that restored it if something pushed *oof* out.

The question is how strong a causal claim the descriptive co-occurrence in Section 2 supports. One proposed local relation connects prosodic isolation and syntactic non-integration: a form produced as its own intonation unit is less available for complementation or modification, while a form outside clause structure is more likely to receive independent prosodic packaging. *Damn!* standing alone has both properties; prenominal *damn* in *the damn car* has neither, because sitting inside a noun phrase costs it the intonational boundary that accompanies the standalone use. Written-corpus proxies don't establish either causal direction, so Table 2 treats the coupling as a candidate rather than a result.

Other proposed processes are interactional or directional. Uptake may affect future use; repeated use may affect expectation; expectation may affect later uptake. *Mm-hm* illustrates the hypothesized loop: it serves as a continuer when recipients treat it as returning the floor, and recurrent use in that slot may entrench that affordance. Loss of referential content makes argument structure less motivated; syntactic deactivation pushes a form toward standalone use; repeated conversational deployment entrenches stance, response, repair, or sequence-management functions. These relations would explain why the diagnostics travel together. The evidence below supports some of them, motivates others, and leaves inventory-level maintenance open.

The historical evidence adds a further point. Semantic bleaching, syntactic deactivation, and onomatopoeic or interactional coinage can all recruit forms into a similar pragmatic profile, but they don't do so by the same route or with the same residues. A bleached invocation (*gee*), a deactivated optative (*damn*), and an onomatopoeic or interactional coinage (*oof*, *mm-hm*) may converge on prosodic independence, supplement function, non-referential or use-conditional meaning, and interactional force while retaining different traces of their sources. The result is a recurrent cross-category configuration whose proposed causal relations have different temporal and explanatory profiles.

If borne out, that structure would warrant the declarations in Section 5. The present evidence isn't enough to establish their held-out success or corrective control, since it doesn't identify a tracked category-level variable, an error signal, or a response pathway that restores the interjectional profile when it's perturbed (cf. Weinberger, 2026).

3.2 Causal networks vs. prototype theory

Prototype theory (Rosch, 1975; Taylor, 2003) was developed for exactly the kind of data Section 2 documents, and it handles that data well: graded membership, better-and-worse exemplars, and fuzzy boundaries via overlapping similarity spaces. It also correctly predicts that boundary disputes (interjections vs. fillers, vs. routine formulae) should recur instead of being resolvable by definition. These are genuine explanatory achievements, and the causal-network account doesn't replace them.

The accounts differ in what they buy that data with. Prototype theory buys it with degrees of membership. The causal-network account doesn't need them, because each field-relative cut is determinate and the disputes concern which cut an inquiry should be using (Section 6.2). That contrast generates a test rather than a present result: apparent gradience should decompose under the conditions that determine the relevant classification rather than persist as irreducible fuzziness.

A causal-network account adds a different kind of question. Prototype theory explains why borderline cases should be expected: categories are organized around better and worse exemplars. It doesn't explain why the properties cluster in this particular pattern or what should happen when one of their links is disrupted.

The difference matters because the causal-network account generates predictions about change, conditioning, and residues. It predicts that forms recruited by different pathways should retain different traces of their source constructions, that appropriate conditioning may decompose some apparent gradience, and that some properties may be more central to network order than others (hub-property asymmetry). Section 6 finds unequal support: residue is the strongest probe, country conditioning is more limited, and hub-property asymmetry remains open.

Pathway residues provide a further discriminator. If the proposed causal links are real, messy boundary cases shouldn't be messy in arbitrary ways. Retained complements should point to verbal sources; euphemistic phonological deformation should point to taboo-managed invocative sources;

regional concentration is compatible with regional or indexical entrenchment but doesn't establish it. Prototype theory offers no analogous retrodictive test tied to independently warranted histories.

3.3 What causal structure adds

Several analytical stances generate different predictions for interjections. The weakest is a property list: interjections share prosodic isolation, non-inflection, non-referentiality, and emotive force. That's descriptively useful, but it makes no prediction about what happens when one property changes.

Feature specifications are stronger alternatives, for example [+peripheral, -argument-taking, +expressive]. This generates some of the same distributional consequences: no inflection follows from the absence of phi-features, and no argument-taking follows from the relevant syntactic feature bundle. But feature specifications treat those consequences as entailments of the feature bundle, not as effects of identifiable processes.

A constructional account is a stronger competitor. It can represent many of the same pairings of form, meaning, and use (Croft, 2001; Goldberg, 1995, 2006). The causal-network claim is compatible with that representational move. It asks a different question: which positions in a constructional inventory are projectible, why do their properties cohere, and which changes should make them come apart? Construction grammar models the inventory; the projectibility question asks where the inventory supports cross-constructional and cross-field inference. The two main pressure points are pathway residues and field-relative decomposition. If those payoffs fail under further testing, the causal-network account collapses to a descriptive inventory rather than an explanatory category analysis.

The causal-network account adds five claims that property lists, feature bundles, and constructional representations don't settle by themselves:

1. Co-movement. Prosodic isolation, supplement function, non-referentiality, and pragmatic force recur in a patterned way. The account asks which links explain why observing one property licenses expectations about another.
2. Asymmetry. Some properties are upstream or more diagnostic than others. Prosodic packaging may make syntactic integration less available, while the reverse relation may have a different status. A flat feature bundle doesn't by itself represent that proposed ordering.
3. Persistence. A category profile may remain available while individual forms enter and leave. Inventory turnover is compatible with that stability claim but doesn't establish it.
4. Boundary behaviour. Different recruitment pathways should leave different residues at the category boundary. Feature theory can list residual properties but doesn't explain why they align with particular histories.
5. Failure. When a prediction fails, the pattern of failure should constrain the corresponding link or declaration. *Fie*'s diagnostics decline over the same broad period without the predicted ordering. That result limits ordered erosion; it doesn't by itself identify simultaneous or community-wide attrition.

If a feature-based account can replicate co-movement, asymmetry, persistence, boundary behaviour, and failure without network order, causal structure is doing no additional explanatory work. The empirical pressure point is whether the probes in Section 6 support those claims.

3.4 Three field-relative categories

The causal-network account developed by Khalidi (2013, 2018) also explains why projectibility is field-relative. If kinds are nodes in causal structures, different fields may track different nodes because they ask different explanatory and predictive questions. The resulting categories need not be domain-pure or merely interest-relative. An inquiry fixes the scope of a declaration; evidence and worldly organization determine whether the resulting classification and projections succeed.

The proper-noun/proper-name distinction supplies a close linguistic parallel. PROPER NOUN is principally useful to grammar, while PROPER NAME is principally useful to semantics, pragmatics, and onomastics. Neither is domain-pure. Proper names have grammatical structure, and proper nouns have conventional referential tendencies. Their extensions still come apart: *the University of Toronto* is a name whose head is a common noun, while a proper noun can be used predicatively or metalinguistically without straightforwardly naming its usual bearer. Neither classification is a defective version of the other.

The interrogative/question distinction makes the same point across grammar and interaction. Syntacticians track INTERROGATIVE, a clause type with properties such as subject–auxiliary inversion and fronting of interrogative words. Pragmatists track QUESTION, an action that requests information, makes an answer relevant, and organizes uptake. The two often coincide and also come apart: *Can you pass the salt?* is interrogative in form and ordinarily a request in use.

Interjections create a similar problem with less terminological help. Syntax, semantics, and interactional pragmatics all use the word "interjection", but each field tracks a slightly different category: different contrasts are preserved and different projections are at stake. The common bearer is the conventionalized sign-type defined in Section 1, so the three classifications apply to objects of the same type. I distinguish them with subscripts:[2]

- INTERJECTION$_{\text{SYN}}$ is the grammar-facing category represented in *CGEL* and Quirk et al. Its principal projective family concerns grammatical distribution and packaging. Admission evidence is primarily morphosyntactic and constructional: characteristic supplement or standalone distribution, morphological invariance in the relevant use, and severe restrictions on ordinary argument structure and integration. This is a lexical category in the grammarian's ordinary sense, without thereby being ontologically prior.
- INTERJECTION$_{\text{SEM}}$ is the semantics-facing category, overlapping with Potts (2007)'s EXPRESSIVES. Its principal projective family concerns conventional meaning contribution. Its profile includes independence from at-issue content, nondisplaceability, perspective dependence, and descriptive ineffability (Potts, 2007, p. 166). Prenominal *damn* in *the damn car* belongs here while falling outside INTERJECTION$_{\text{SYN}}$.
- INTERJECTION$_{\text{PRAG}}$ is the interaction-facing category, overlapping with Stivers (2019)'s response tokens and Dingemanse (2020)'s liminal signs. Its principal projective family concerns conventional interactional action and sequential organization. Minimal response, stance, repair,

[2] The interrogative/question contrast uses different words for the syntactic and pragmatic categories because both terms are established in their respective traditions. For interjections, every field uses the same word, so subscripts are the honest notation. An alternative would have been to adopt distinct terms from the relevant literatures: EXPRESSIVE from Potts (2007) for the semantics-facing category, and response token from Stivers (2019) or LIMINAL SIGN from Dingemanse (2020) for the interaction-facing category.

The Latin *ejaculatio*, a throwing out, mirroring *interjicere*, "to throw between" (Libert, 2019), would have been etymologically perfect for the interactional-pragmatic category, but modern English has rendered the term unavailable.

continuer, and change-of-state uses populate this region. *Mm-hm* belongs here while its INTERJECTION$_{SYN}$ status remains disputed.

The subscripts identify principal consumers and projective families, not the exclusive provenance of the categories' properties. Each classification may depend on relations among form, grammar, meaning, use, convention, and history. The three needn't be equally rich or symmetrical.

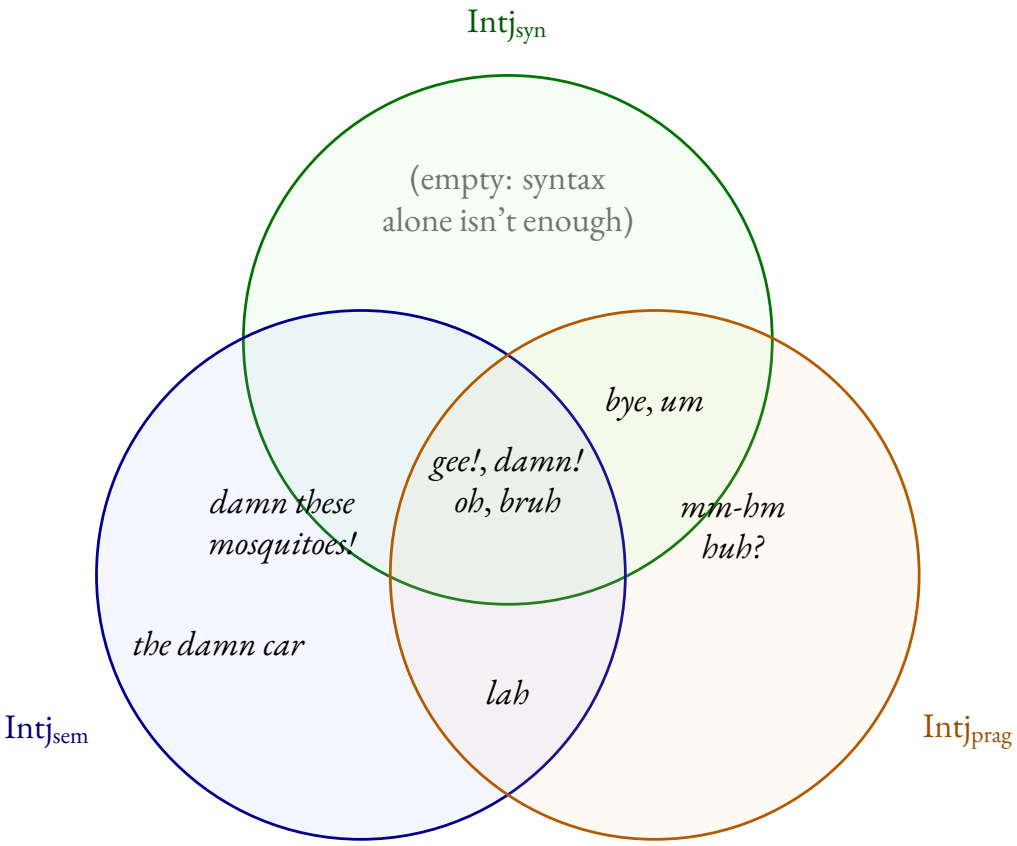


Figure 2: The three field-relative categories overlap in extension but come apart, and where they come apart is evidence that they're three. Placement follows Table 1; proximity to a boundary marks a qualified or disputed classification, not an additional graded-membership claim. Prenominal *damn* carries the semantics-facing category without the grammar-facing one, *mm-hm* the interaction-facing without the semantics-facing, and *bye* and *um* neither the semantics-facing category nor undisputed interaction-facing standing. Malaysian and Singaporean *lah* carries the semantics- and interaction-facing categories without the grammar-facing one, for the reason given in Section 4.5.

The subscript notation makes explicit that these are overlapping but non-identical categories, each associated with different proposed projections and causal relations. Bare "interjection" is a conventional cover term, not a fourth category. Canonical sign-types belong to all three, but the intersection has no standing or projective force beyond the component classifications. The remaining sections analyze their relations rather than trying to vindicate the overlap as another category.

Table 2: Status of the proposed ordering and stabilizing links

Link	Status	Evidence type	Main vulnerability
Semantic bleaching from invocatives	Historically documented	Gehweiler-style corpus trajectory	Operationalizing invocation versus exclamation.
Syntactic deactivation from verbs, optatives, and imperatives	Plausible and reconstructive	Residual complement structure, optative and imperative frames, discourse-marker uses	Needs fuller diachronic series.
Onomatopoeic or liminal coinage	Plausible entry route	Synchronic profile of coined forms	Coinage isn't the same kind of process as bleaching.
Prosodic-syntactic coupling	Stabilizer candidate, partly proxy-based	Supplement syntax, punctuation, standalone position	Written corpora underrepresent prosody.
Non-referentiality ratchet	Proposed ordering link	Argument-structure restrictions	*Yes*, *no*, and response tokens complicate the property.
Routine entrenchment	Stabilizer candidate for INTERJECTION$_{PRAG}$	Conversation-analytic and pragmatic literature	Needs more spoken-interaction data for these cases.
Regional/indexical conditioning	Maintenance hypothesis	Regional concentration and register sensitivity	Identity-work evidence remains outside this dataset.

4 Causal structure: recruitment and coupling

The property cluster documented in Section 2 raises two questions that co-occurrence alone can't answer: how do items enter the interjectional region of the network, and which causal links make the properties travel together? In what follows, "the interjection cluster" refers to the broad property profile. I use "the intersection" only for cases that belong simultaneously to INTERJECTION$_{SYN}$, INTERJECTION$_{SEM}$, and INTERJECTION$_{PRAG}$. A causal-network account requires both a diachronic account of recruitment and testable proposals about synchronic order among properties.

Table 2 sets out each proposed link with its current evidential status, the kind of evidence that would bear on it, and the main way it could fail.

4.1 Recruitment pathways

Section 2 already pointed to three recruitment pathways: semantic bleaching from nouns, syntactic deactivation from verbs, and nonce coinage via onomatopoeia. The first is historically demonstrated; the second and third are proposed pathways with clear examples but without the extended diachronic documentation that Path 1 has. The first two are analysed in detail here; the third is taken up in Section 4.7.

The pragmatic issue is how changes in communicative use become categorially consequential: grammaticalization and pragmaticalization matter here because they alter a form's communicative job as well as its position in a syntactic tree (cf. Heine & Kuteva, 2002; Traugott & Dasher, 2002).

4.2 Path 1: religious noun to interjection

Gehweiler (2008) traces the development of *gee* from *Jesus* through three positions:

(5) a. Proper name used as invocation (*Jesus!* addressed to the deity)
b. Secondary interjection: exclamatory use, no longer addressing the deity (*Jesus!* as an expression of surprise)

c. Primary interjection: phonological modification yields *Jeez* and *Gee*, and *gee!* detaches from the source noun

The two transitions don't work the same way, and Gehweiler is emphatic about the difference. She explains the first as gradual and speaker-based, through Traugott and Dasher's invited inferencing, and "advanced by the syntactic isolation of *Jesus*" (Gehweiler, 2008, p. 85). The second she treats as abrupt and hearer-based: tabooistic clipping produced *gee!*, and she insists that this step "was not gradual" and "can therefore not be explained with reference to Invited Inferencing or syntactic position" (Gehweiler, 2008, p. 88). The positions layer cluster properties unevenly. Prosodic isolation is already present at the invocation, because a vocative like *Jesus!* occupies its own intonation unit (contrast the integrated argument in *Put your faith in Jesus*). What changes is the content: non-referentiality arrives with the secondary interjection, when the speaker is no longer addressing the deity. Phonological divergence then makes the etymological connection opaque, leaving a standalone interjection that many speakers don't connect to its source. For the first transition the vocative is the gateway, and what it supplies is syntactic: an utterance-initial slot set apart from the rest of the utterance, which is the isolation Gehweiler credits with enabling the categorial change (Gehweiler, 2008, p. 85). She denies that the same explanation extends to the second.

That discontinuity is an asset for the path-dependence claim rather than an embarrassment for it. A gradual slide would leave a form with no particular mark of where it came from; an abrupt reanalysis leaves exactly the kind of residue the account needs, since the phonological opacity of *gee!* is a permanent trace of the clipping that produced it. The first transition is constrained in the way Gehweiler describes: once the invocation loses its referential force, the phonological connection to the source is the last link to the nominal category, and weakening that link by phonological modification enables full independence. Gehweiler's corpus data documents the constraint: the phonological modifications (*Jeez*, *Gee*, *Gosh*, *Golly*) cluster in the seventeenth and eighteenth centuries, with the transition from invocation to primary interjection happening relatively rapidly once the modifications appear (Gehweiler, 2008, pp. 80–85).

Two cautions matter for the general claim. The boundary between invocation and exclamation is hard to operationalize in historical texts, since prosodic and interactional evidence is indirect. The seventeenth- and eighteenth-century clustering of euphemistic forms may reflect external taboo pressure as well as internal category pressures. The causal-network claim doesn't require Gehweiler's sequence to be an autonomous internal cline. It requires the weaker and more historically plausible claim that social pressure, invited inference, and reanalysis can jointly push a source noun toward the same interjectional profile.

Two processes operate within this path. Gehweiler (2008, p. 76) distinguishes gradual Invited Inferencing (speaker-based: the expressive use is conversationally implicated, then conventionalized) from abrupt reanalysis (hearer-based: the hearer parses the exclamation as a standalone unit rather than an invocation). Both feed the same cluster, but they operate at different timescales and involve different agents. Their convergence shows that the same present-day classification can be reached by different routes; it doesn't by itself establish synchronic network order.

The first transition generates a specific prediction: partial bleaching should be detectable by the absence of downstream properties. A form that has reached the secondary interjection but no further should be non-referential, retain the source phonology, and resist full prosodic independence. The prediction is testable and goes beyond historical description.

4.3 Path 2: verb to interjection

A second proposed trajectory enters through syntactic deactivation rather than phonological modification. *Damn* has a verbal source, and the optative construction supplies a plausible gateway. Optatives include *God save the Queen* and *God help you*, where a main-clause subjunctive survives in a narrow range of formulaic frames (Huddleston & Pullum, 2002, p. 944). *God damn him* has that shape, and the wish interpretation of *damn him* is compatible with reanalysis after omission of the subject.

The proposed path is reconstructive rather than a demonstrated token-by-token derivation. The optative already contains plain-form *damn*, so no tense morphology is lost in the proposed transition. What survives in *damn these mosquitoes!* is an NP complement compatible with verbal source syntax. In *CGEL*'s synchronic analysis the construction is headed by an IntjP rather than a VP (Huddleston & Pullum, 2002, 1361fn), so its complement-taking qualification has to remain visible.

If the reconstruction is right, the process is decategorialization involving reduced verbal argument structure and increasing conventional expressive force. As a verb, *damn* contributes content involving condemnation. Standalone *damn!* conventionally displays a stance of frustration. The present-day contrast and complement residue motivate the path, but a fuller diachronic series would be needed to establish its sequence.

This doesn't count as degrammaticalization in the strict sense, since the form doesn't gain the morphosyntactic substance of a major lexical category (cf. Norde, 2009). It's better treated as pragmatic conventionalization with syntactic deactivation: the verbal source loses argument-structural obligations as the interjectional use gains procedural and interpersonal force.

For other verb-derived forms the imperative is a candidate gateway. Imperatives characteristically lack an overt subject and can stand alone prosodically. Discourse-marker uses of *come on* and *look* show reduced argument-structural obligations in the relevant sign-types. Those synchronic patterns are compatible with deactivation, but they don't by themselves establish a historical series.

The proposed contrast generates a testable residue prediction: sign-types that retain complementation should retain more source-compatible grammatical behaviour than sign-types with no residual argument structure. The present examples motivate that prediction; they don't yet establish a category-wide correlation.

4.4 Convergent pathways

The two paths enter the cluster from different directions, one through phonological modification of a bleached noun, the other through syntactic deactivation of a verb, but both exploit a gateway construction that already provides standalone or peripheral packaging: the vocative for Path 1, and, on the reconstruction above, the optative or imperative for Path 2. Both can yield similar present-day profiles. The documented strength is asymmetric: the invocative history is established in substantially more detail than the proposed verbal path.

Both paths can be situated within Traugott and Dasher (2002)'s subjectification trajectory (content > procedural, nonsubjective > intersubjective), but they instantiate it differently. Path 1 is grammaticalization in Traugott's technical sense: the invocation carries content meaning (addressing an entity) plus subjective stance (reverence); the secondary interjection loses content and gains procedural meaning (expressive force); phonological modification completes the shift to fully procedural, intersubjective meaning, with hearers interpreting the form as a conventional expression, not an invocation. Gehweiler's own verdict on that last step is worth keeping in view: because it turns on

hearer re-analysis rather than speaker-invited inference, it's the one transition her account doesn't treat as subjectification (Gehweiler, 2008, p. 88). Path 2, as described above, is decategorialization: loss of verbal morphosyntax with gain of expressive force.

Onomatopoeia (Section 4.7) is categorically different. Nonce coinages enter the cluster with no prior lexical history (no bleaching, no deactivation, no subjectification). Three diachronically distinct processes (grammaticalization, pragmaticalization, nonce coinage) can yield sign-types assigned to one or more of the same synchronic categories. Grammaticalization theory alone doesn't subsume all three. A causal-network account offers a common representation of the recurrent outcomes and retains the distinct routes. Whether all are linked by the same synchronic ordering relations remains an empirical question.

4.5 Synchronic coupling

Recruitment hypotheses concern how items enter the cluster. The synchronic question is whether the cluster has network order now, and what might keep that order stable. Four proposed links are relevant: prosodic-syntactic coupling, a non-referentiality ratchet, interactional routine entrenchment, and regional/indexical conditioning. They don't have the same status. The first two order structural and semantic diagnostics. The third is the central stabilizer candidate for INTERJECTION$_{\text{PRAG}}$. The fourth remains a maintenance hypothesis.

4.5.1 Prosodic isolation and syntactic non-integration

These two properties reinforce each other, but the evidence is asymmetric. Prosodic separation plausibly makes syntactic integration harder, since a form in its own intonation unit is less available as a complement or modifier. But the corpora used below are written, and prosodic isolation is tracked there through proxies (Section 2.1), so they can't establish that direction.

The diachronic record does support the syntactic half. Gehweiler (2008, p. 85) finds that invocational *Jesus* "typically stands utterance-initially and – like other interjections – set apart from the rest of the utterance, or even on its own", and argues that "only because of this syntactic isolation of the proper name – and Invited Inferencing, of course – could the secondary interjection *Jesus!* develop from the proper name". That's evidence that syntactic isolation, together with invited inferencing, enabled a categorial change, not that prosody enabled the syntax. She also limits it: as Section 4.2 noted, she takes the step from *Jesus!* to *gee!* to lie outside both invited inferencing and syntactic position (Gehweiler, 2008, p. 88).

The same coupling is visible synchronically. Standalone *Damn!* has both properties: it occupies its own intonation unit and has no role in clause structure. In *I think that, damn, we're going to be late*, *damn* remains possible only as a parenthetical supplement, set off from the *that*-clause. In *the damn car*, the form loses both properties at once: it's inside the noun phrase and receives the prosody of the phrase.

The prediction is concrete. If a novel form is reliably packaged as its own intonation unit, it should resist complementation, modification by ordinary phrase-internal material, and coordination with integrated phrases. If it's pulled into a host phrase, prosodic isolation should weaken together with INTERJECTION$_{\text{SYN}}$ status.

4.5.2 The non-referentiality ratchet

Bleaching removes referential content, which weakens the semantic motivation for argument structure. A form with no entity or event to construe is less available as an argument, complement, or

modifier. Once that syntactic role is lost, prosodic isolation becomes easier to keep in place. The result is a one-directional cascade. Once argument structure is lost, the form occupies a different functional niche, so ordinary pressures toward referential syntax are reduced.

This ratchet supports a different set of predictions from the prosodic-syntactic link. A non-referential stance form should resist definiteness, quantification, and argument position: **the damn!*, **every ouch*, **Damn surprised me*. It should also resist embedding as propositional content (**She said that damn*) unless it's quoted as a form. If referential content becomes active again, the form should move toward noun, verb, or routine-formula regions of the network. The ratchet keeps the form near the interjectional region by blocking return to the source category, while prosodic-syntactic coupling links the central structural properties.

4.5.3 Interactional routine entrenchment

The interaction-facing category needs a candidate stabilizer of its own. Conversation repeatedly places interjectional forms in recognizable sequential slots: continuers, repair initiators, change-of-state tokens, stance displays, and response tokens (Dingemanse, 2020; Stivers, 2019). Dingemanse (2024) argues that these interactional uses are central rather than peripheral for high-frequency interjections.

Repeated deployment in these slots favours a particular package of properties. Two levels are in play here, and the argument needs both. On a given occasion, a continuer such as *mm-hm* works because it can be recognized without taking over the turn or adding propositional content. A change-of-state token such as *oh* works because it can mark an information update before the speaker builds further at-issue content. A stance token such as *bruh* works because it can display evaluation while leaving the surrounding proposition untouched. Across many occasions, that occasion-level fitness is supposed to accumulate, so that the interactional routine favours short, standalone, prosodically independent, non-referential material in the inventory at large. The accumulation is an assumption here rather than a finding. The probes in Section 6 bear on the inventory-level pattern, not on the step from one level to the other.

If borne out, this candidate stabilizer would support the projectibility of INTERJECTION$_{\text{PRAG}}$. If a novel form is recruited as a minimal stance, response, repair, or backchannel resource, one should expect turn-initial or standalone availability, independent prosodic packaging, resistance to embedding under reported speech, and availability for uptake as an interactional move. The same candidate could also explain why the function can persist when category membership weakens: a full clause such as *I agree* can do agreement work, but it lacks the reduced, non-propositional packaging that makes *mm-hm* a central INTERJECTION$_{\text{PRAG}}$ case.

4.5.4 Regional/indexical conditioning

Some interjections are sharply conditioned by region, register, or community, and some may also do identity work. *Lah* in Singapore and Malaysia, *yaar* in India and Pakistan, *haba* in Nigeria, and *bruh* in African American English (Louf et al., 2023) are plausible cases.

But they aren't the same kind of case, and the difference matters for which node the conditioning operates on. *Haba* is free-standing: Unuabonah and Daniel (2020) avoid the terms utterance-initial and utterance-final for it and its Nigerian congeners "because they are free standing and are far more detached from the utterances than PMs" (Unuabonah & Daniel, 2020, p. 69). *Lah* is clause-final and prosodically dependent on its host, and this holds on either current account of it:

Lim (2007, p. 447) reports that Gupta takes the particles to "function as the carrier of the intonation contour of the utterance", while the rival analysis gives them lexically-specified tone inherited from a substrate. That rival is weakest for *lah* in particular, since *lah* is one of the few particles Lim (2007, p. 463) doesn't match to Cantonese, and its Bazaar Malay and Hokkien sources have "neutral tone and weak stress". Either way *lah* fails prosodic isolation, so it satisfies INTERJECTION$_{SEM}$ and INTERJECTION$_{PRAG}$ without INTERJECTION$_{SYN}$ (Figure 2). Regional conditioning can in principle recruit a sign-type to the interaction-facing category without recruiting it to the grammar-facing one, a pattern compatible with *lah* but not *haba*. The GloWbE evidence in Section 6, modelled in Appendix A, shows additional country-level variation in overall item frequencies. It doesn't by itself establish social indexing in the stronger sociolinguistic sense: uptake, stance, enregisterment, style shifting, metapragmatic awareness, or distribution by speaker group within comparable contexts.

Regional/indexical conditioning is a maintenance hypothesis. The open question is whether it stabilizes the cluster profile (property co-occurrence) or individual items (inventory persistence). If it stabilizes the profile, regional/indexical conditioning helps constitute the node. If it stabilizes only the item, it operates as a lexical-persistence process alongside the interjection cluster.

When a regional interjection is adopted by out-group speakers as ironic quotation, the causal-network account predicts a contrast between inventory persistence and profile stabilization. If out-group *lah* loses prosodic isolation or emotive force while retaining its social-marking function, regional/indexical conditioning stabilizes the inventory alone. If it retains the full profile, the link is part of the causal structure of the category. This comparison remains for future work.

The account proposes two structural ordering links (prosodic-syntactic coupling and the non-referentiality ratchet), one interactional-pragmatic stabilizer candidate (routine entrenchment), and one regional/indexical maintenance hypothesis. If correct, the first three would help explain why prosodic, syntactic, semantic, and pragmatic diagnostics travel together synchronically. Their unequal evidential status has to remain visible in the probes and conclusion.

Figure 3 summarizes the proposed causal network. It's a schematic map of the links used in the argument rather than a complete model of every dependency among the diagnostics.

4.6 Causal residues at the category boundary

The proposed interjection network is a convergence region, but convergence needn't erase history. Forms recruited through different pathways may share parts of a present-day profile while retaining traces of their source constructions. Where the history is independently documented, those traces can distinguish pathway signatures from otherwise unexplained irregularity.

This supplies a second kind of proposed inference. Section 5 states forward declarations from observed to unobserved properties. Historical pragmatics also permits a retrodictive counterpart: a residual property can warrant an inference about its source pathway when the historical evidence is independent rather than reconstructed from the same residue. A causal-network account represents that relation explicitly.

Table 3 collects candidate residues and the retrodictive inferences they motivate at different evidential strengths.

The residue analysis turns apparent exceptions into candidate evidence whose strength depends on the history available. Complement-taking *damn these mosquitoes!* is compatible with a verbal path; *gee* securely records taboo-managed invocative recruitment through euphemistic phonology. *Lah* motivates a regional or indexical maintenance hypothesis but doesn't establish one. Prototype

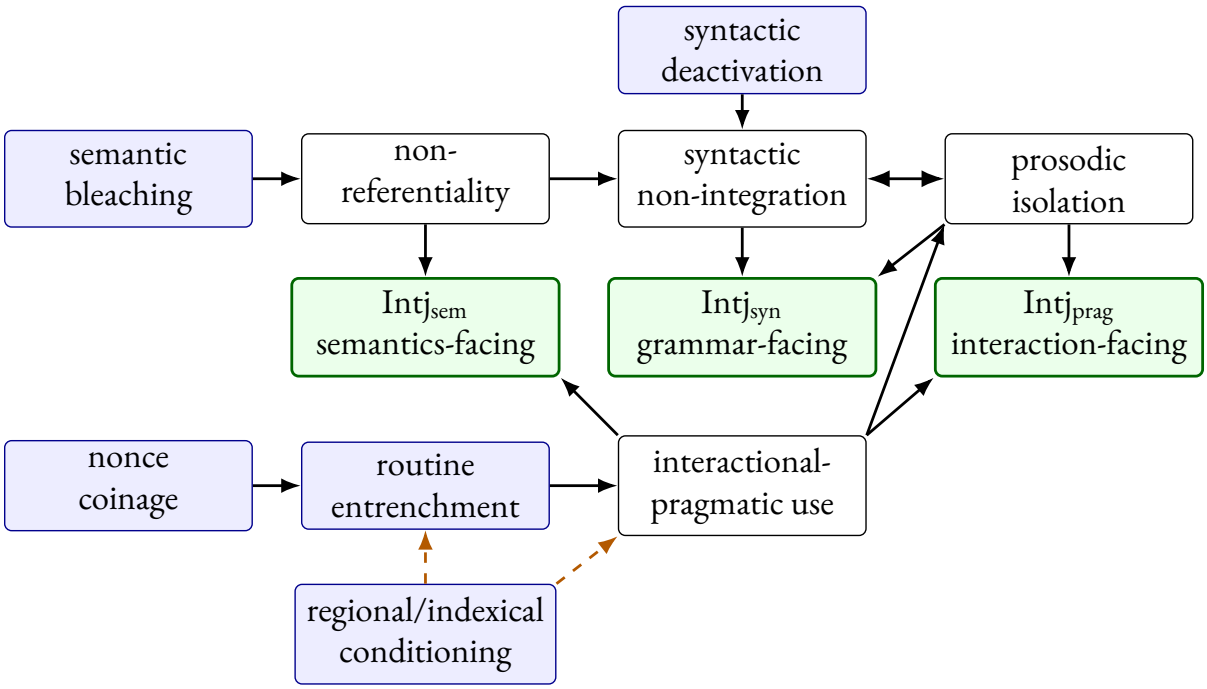


Figure 3: Schematic causal network for English interjections. Blue nodes mark recruitment, ordering, or stabilizing processes; white nodes mark diagnostics; and green boxes mark field-relative interjection categories. Solid arrows indicate proposed ordering links. The double arrow marks reciprocal prosodic-syntactic coupling. Dashed orange arrows mark the regional/indexical maintenance hypothesis. The three field-relative categories overlap in extension and generate different prospective declarations.

Table 3: Pathway residues in the English interjection cluster

Pathway	Gateway	Residue	Retrodictive prediction
Religious noun or proper name	Vocative invocation	Taboo avoidance, euphemistic phonological divergence, loss of reference without complement residue (*Jesus!* > *jeez*, *gee*; *God!* > *gosh*, *golly*)	Euphemistic deformation plus no argument structure points to an invocative source.
Verb	Optative or imperative frame	Complement residue from the optative (*damn these mosquitoes!*); second-person orientation, directive force, and particle retention from the imperative (*come on*, *look*)	An NP complement is compatible with a verbal or optative source, and a directive orientation with an imperative one; deciding between reconstructions requires independent history.
Onomatopoeic or interactional coinage	Iconic or interactional vocalization	Atypical phonology, reduplication, spelling variation, weak etymological transparency (*ugh*, *whew*, *tut-tut*, *uh-oh*, *shh*)	Noncanonical phonology without a plausible lexical source points to iconic coinage.
Interactional recruitment or regional/indexical conditioning	Sequential slot or community marker	Regional skew, register sensitivity, indexical persistence (*lah*, *yaar*, *haba*, *bruh*, *mm-hm*)	Uneven distribution by community motivates a regional/indexical entrenchment hypothesis but doesn't establish the historical route.

theory predicts fuzzy boundaries. It doesn't by itself predict that different boundary failures should align with independently documented historical routes.

Cuenca (2002, p. 40) reaches the same claim from the grammaticalization side, observing that secondary interjections "tend to become syntactically and morphologically invariable, but whenever the process is not accomplished some syntactic features can persist", her example being *Good for you/her*. She gets there from subjectification rather than from a causal network, so two independent routes reach the claim that unfinished recruitment leaves syntactic traces.

The same logic applies when forms leave the cluster. A preregistered COHA check selected overt derivatives of *wow*, *boo*, *ooh*, and *shoo*, then examined their included VP-head uses. In the earliest confirmatory windows, *wowed*, *booed*, *oohed*, and *shooed* retain source-interjection semantics (Appendix B). Because derivative form was selected in advance and VP-head syntax was part of the inclusion criterion, the result is semantic residue conditional on those derivative uses, not a general

finding that interjections gain morphology or syntax.[3] These exit cases fit the residue account: in the selected integrated uses, source-interjection semantics persists. They probe boundary behaviour, not corrective control.

The *fie* study marks the limit of residue reasoning under global attrition. The pre-registered prediction was that INTERJECTION$_{\text{PRAG}}$ would weaken before INTERJECTION$_{\text{SEM}}$, and INTERJECTION$_{\text{SEM}}$ before INTERJECTION$_{\text{SYN}}$. In COHA, all three node diagnostics decline over the same broad period, and the predicted order receives low posterior support (Figure 4).[4]

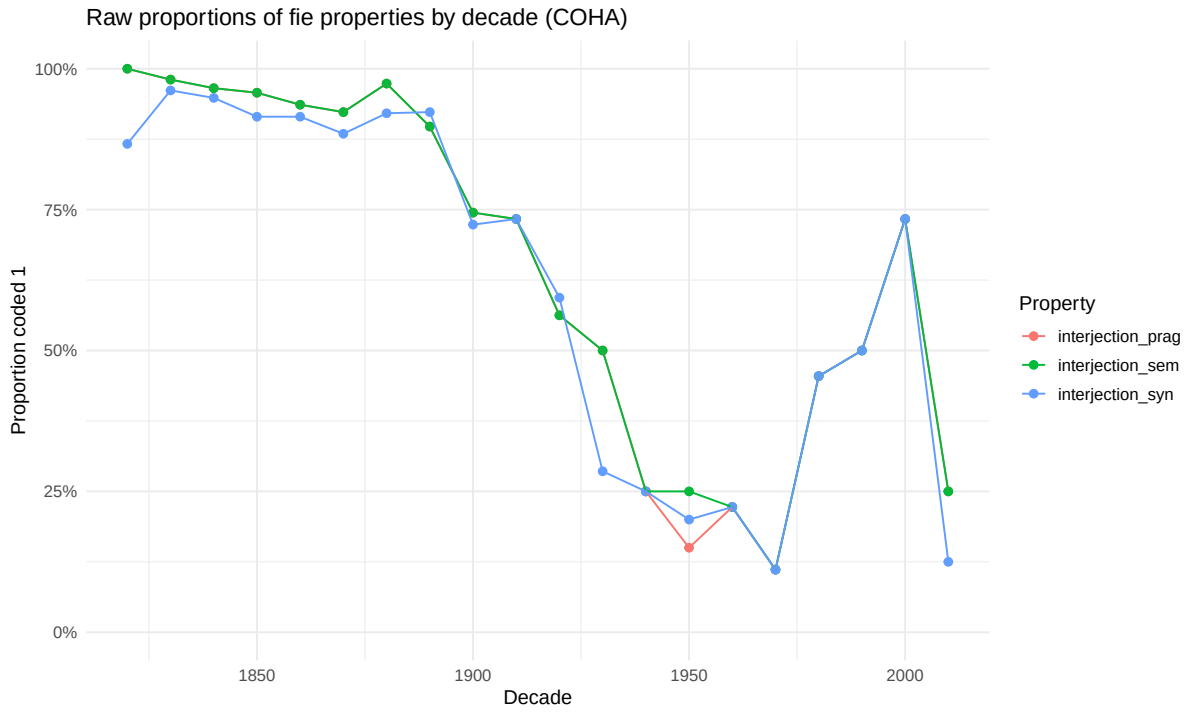


Figure 4: The prediction that failed. Proportion of *fie* tokens coded 1 for each diagnostic, by decade (COHA, 1820–2019; $n = 616$ coded tokens). The pre-registered prediction was that INTERJECTION$_{\text{PRAG}}$ would weaken first and INTERJECTION$_{\text{SYN}}$ last. Instead all three decline over the same broad period. The result doesn't support the predicted ordering; the marginal trajectories alone don't establish equality or simultaneity. Full model and coding in Appendix C.

The result is compatible with community-wide attrition affecting several diagnostics over the same period, but it doesn't identify that process or establish simultaneous loss. More securely, it supplies a failure condition: the proposed ordering isn't visible in this case. Interjections may carry traces of recruitment, but attrition need not expose those traces in an ordered sequence.

[3] The analysis plan was committed before extraction and coding (`f1a2193`, 27 May 2026). The full pre-registration, data, and code are archived at https://github.com/BrettRey/interjections-hpc.

[4] The analysis plan was committed before data extraction (`e18675b`, 15 March 2026). The full pre-registration, data, and code are archived at https://github.com/BrettRey/interjections-hpc.

4.7 Inventory turnover

Pathway residues bear on how interjectional forms entered the network. Turnover asks the complementary descriptive question: how does the lexical inventory change? Inventory replacement is compatible with a persistent category position, but it doesn't by itself establish profile stability or projectibility.

The Corpus of Historical American English (COHA) illustrates this turnover. In the 1820s the six most frequent interjection lemmas are *no*, *oh*, *yes*, *nay*, *o*, and *ah*. By the 2010s half that list has gone: *nay*, *o*, and *ah* are replaced by *yeah*, *hey*, and *uh*, while *no*, *oh*, and *yes* hold their places throughout (Figure 5). *Nay* falls from 175 tokens per million to 2.2; *yeah*, absent from the 1820s altogether, rises from nothing to 723.[5] The tagged inventory changes substantially across the corpus period.

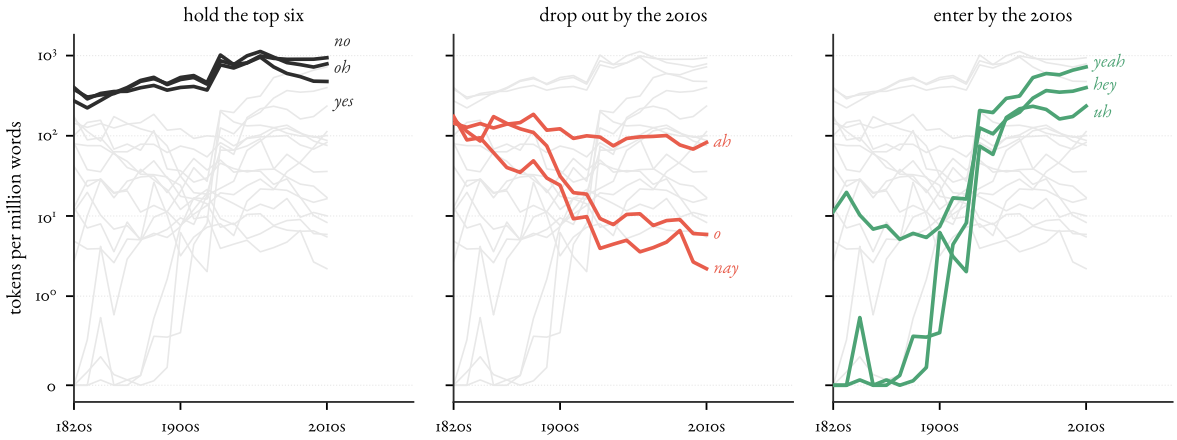


Figure 5: Turnover in the COHA interjection-tag inventory. Each panel shows all twenty of the most frequent UH-tagged lemmas in COHA (grey), with one group picked out. Half the 1820s top six is gone by the 2010s, and the forms that replace it were absent or negligible when the corpus starts. Tokens per million words on a symmetric log scale, so that forms at zero remain visible. The tag inventory isn't an independent test of category stability. Extracted 2026-07-27 and archived under `analysis/hand-extracted/`.

Regional variation supplies a synchronic comparison among local inventories. The GloWbE corpus (Davies & Fuchs, 2015, 1.9 billion words across 20 English-speaking countries) shows that candidate interjectional forms are distributed unevenly across country-labelled web sections, with *lah* almost exclusively in Singapore and Malaysia, *yaar* in India and Pakistan, and *haba* in Nigeria. The raw-string concentrations support a descriptive geographic claim, subject to the measurement limits in Section 6; they don't establish sociolinguistic conditioning.

[5] Part-of-speech–tagged frequencies in COHA (Davies, 2010), extracted 2026-07-27 and archived in `analysis/hand-extracted/` as `nay-yeah-coha-by-decade.csv` and `interjection-lemmas-coha-by-decade.csv`. The 1820s ranking is taken from the complete tagged list for that decade, 111 forms, so it doesn't depend on the all-corpus top twenty; *farewell*, *yea*, *ay* and *lo* are frequent early and absent from that top twenty, and none of them displaces the six named here. Frequencies are normalized per million words. The interjection tag captures a conventional inventory rather than a theoretically motivated one; the point here is the turnover pattern, not the tagging decisions.

Onomatopoeic coinage is also a proposed source of new interjectional sign-types. Two clarifications, because ONOMATOPOEIA is often handled as though it named a category alongside noun and verb. It doesn't. It names a relation between a form and a sound, and forms standing in that relation are spread across categories: *bang* is a noun in *a loud bang*, a verb in *he banged the door*, and an interjection in *Bang! You're dead*. So the claim isn't that a category of onomatopoeias gets absorbed into the interjections. It's that imitation is one of the routes by which new forms arrive at the interjectional profile.

Nor is the route closed in principle. The stock of imitable sounds isn't fixed by the stock of animals: devices supply *beep*, *ping*, and *whirr*, and any sound can be re-rendered, which is how comics gave English *bam* and *pow* beside older forms. Nonce interjections can be freely coined (Quirk et al., 1985, p. 74). Which of the three categories a coinage enters, and which properties it acquires with conventionalization, are empirical questions rather than automatic consequences of imitation.

This generates a testable prediction: a newly conventionalized interjectional use may acquire several parts of the recurrent profile without traversing the documented noun-to-interjection or proposed verb-to-interjection paths. Take *bruh*. *Oxford English Dictionary* (2026a) records it as a noun from 1894, as a title before a man's name, and from 1967 as a familiar form of address, where every quotation is vocative (*Hey, bruh*, *Be cool, bruh*). The standalone use addressed to nobody is later, and the entry doesn't record it as an interjection at all. Its rise shows up in COCA, from 0.01 per million in 1990–94 to 1.09 in 2015–19.[6] That use is prosodically isolated, non-inflecting, non-referential, and syntactically non-integrated, even though the same word family includes the nominal lexemes just described. It illustrates a candidate recruitment case; the corpus counts alone don't establish persistence of a category position.

Dingemanse (2020) characterizes such items as liminal signs, signs at the boundary between sound and speech, whose liminality is functional, not defective. Cultural evolutionary processes "sample the landscape of possible resources" (Dingemanse, 2020, p. 191). The causal-network proposal describes that landscape as a property space whose structure may reflect the recruitment pathways and synchronic links above.

The descriptive cluster, recruitment evidence, and candidate links now make the paper's inferential question precise: what should partial evidence license, and what would establish that it does?

5 Projectibility as a testable hypothesis

Section 4.6 treated retrodictive inference from residue to an independently supported recruitment history. Forward projectibility has a different direction: from a restricted set of observations about a sign-type to a property or environment not yet observed. This section states those declarations and their failure conditions. It doesn't report a direct held-out evaluation.

The distinction matters because three questions can otherwise collapse. A category may be independently well motivated; a causal account may explain some of its internal relations; and its classification may still fail to improve prediction of new cases. Conversely, failure of one declaration wouldn't abolish all three categories. The relevant claims have to be tested and demoted separately.

[6] Tokens sit overwhelmingly in television and film (1.44 per million) and are absent from the spoken section, so the record is scripted representation of speech rather than recorded speech. COHA has the form absent before the 1980s and nine tokens in all. Figures are raw string frequencies; see `analysis/hand-extracted/`.

5.1 Partial-observation inference

Projectibility becomes circular if it says only this: if X is an interjection, it has interjection properties. That's trivially true of any category. A legitimate declaration fixes the source evidence for a particular case and withholds its target. The target can belong to the category's broader individuation profile, but it can't be one of the observations used to classify that case.

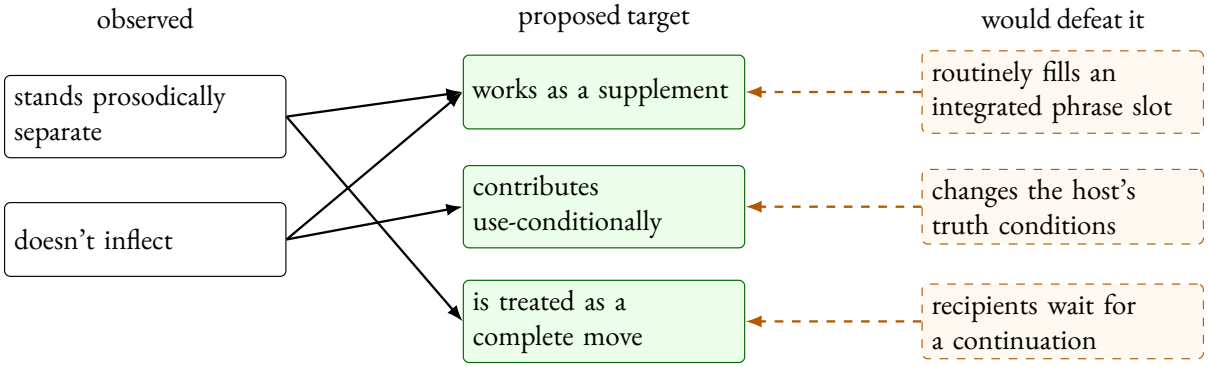


Figure 6: Three proposed projections for *gee*. Two observations form the restricted source evidence; each target is withheld and paired with a defeating result. The diagram states declarations rather than reporting that their held-out success has been measured.

Consider the recently conventionalized standalone use of *bruh*. Standalone position, prosodic separation, morphological bareness, and stance meaning are useful observations, but they can't all be counted first as admission evidence and then as predictions. A declaration has to specify which classification was independently assigned, which observations supported that assignment in this case, and which new environment or property is withheld. Different declarations may use different subsets of the same category's wider diagnostic profile.

The category label adds no information after the complete vector from which it was deterministically computed. That isn't a sensible test of category utility. The relevant comparison asks whether a category- or network-structured representation generalizes from restricted, noisy, or missing evidence to independently measured targets better than an appropriate weaker representation.

Analyst and participant claims also have to be separated. Category-driven projection is the analyst's epistemic use of a classification. Cue-driven inference would be a claim about what speakers or hearers anticipate from partial signals. The category label need not occur in a participant's psychology or cause the target behaviour. The present paper specifies analyst-facing declarations. Whether participants make corresponding online inferences requires separate psycholinguistic or interactional evidence and remains open.

5.2 What each node projects

The three categories do their work as different packages of proposed inference, not as rival definitions. Table 4 fixes one non-trivial declaration for each. The source column identifies the restricted case-level evidence; it isn't an exhaustive definition of the category.

Table 4: Principal prospective projective declarations

Category	Restricted source evidence	Withheld target	Defeating result
Intj_{syn}	Classification established in observed grammatical frames, excluding the target frame	Integration, complementation, coordination, or supplement behaviour in an unobserved frame	No useful held-out advantage over a component-property or supplement-function baseline.
Intj_{sem}	Use-conditional expressive classification, excluding immediacy and repetition behaviour	Immediacy and intensifying rather than redundant repetition (Potts, 2007, p. 167)	Delayed or descriptive contribution, or repetition that produces ordinary redundancy.
$\text{Intj}_{\text{prag}}$	Conventional interactional classification in a source environment, excluding the target sequence	Recipient uptake, next-turn treatment, return of the floor, or flexibility in a withheld sequential environment	Routine treatment as incomplete, or no held-out advantage over a category-free interactional baseline.

These declarations need not be equally strong. Many $\text{INTERJECTION}_{\text{SYN}}$ targets may follow almost directly from observed morphosyntax and be available to any framework that represents the component properties. The semantic and interactional targets reach farther, but that makes their empirical burden greater rather than their success automatic.

Cross-category declarations are possible too: independently assigned $\text{INTERJECTION}_{\text{SYN}}$ may warrant an expectation about $\text{INTERJECTION}_{\text{SEM}}$ or $\text{INTERJECTION}_{\text{PRAG}}$, and conversely. There are six directed relations, they needn't be symmetrical, and the paper doesn't assume that all succeed. Each retained edge requires source evidence that excludes its target and a failure rule. The three-way overlap adds nothing further: canonical co-membership merely makes the successful component projections jointly available.

The three constructions of *damn* show the same point inside a single word family.

Standalone *damn!* belongs to all three categories: the form is prosodically isolated, syntactically non-integrated, use-conditional, and available as a stance marker. For *damn these mosquitoes!*, the picture is mixed: its $\text{INTERJECTION}_{\text{SYN}}$ classification is qualified because *CGEL* treats it as heading an IntjP with limited complement-taking; $\text{INTERJECTION}_{\text{SEM}}$ holds because the meaning is expressive. Table 1 doesn't assign $\text{INTERJECTION}_{\text{PRAG}}$ because this sign-type isn't evidenced as a minimal backchannel or turn-management resource.

For prenominal *damn* in *the damn car*, the categories diverge further: $\text{INTERJECTION}_{\text{SYN}}$ fails entirely (the form is syntactically integrated inside a noun phrase), but $\text{INTERJECTION}_{\text{SEM}}$ holds (use-conditional, perspective-dependent). Neither *CGEL*'s syntax nor Potts's semantics needs to reproduce the other's extension. The three-category analysis represents the divergence without forcing a single verdict across inquiries.

5.3 What would establish the forward claim

A direct evaluation should hold out an entire lexical form family or constructional family, not random tokens from an already familiar item–variety combination. Source classifications have to be assigned without access to target outcomes. The comparison isn't category membership added after its complete diagnostic vector. The test instead compares structured generalization from restricted evidence with a category-free local-cue model of comparable complexity and with the target base rate.

For interactional targets, a spoken corpus can supply independently measured next-turn response, recipient uptake, and floor-transfer trajectories. For semantic repeatability, a separate judgement or production design may be needed. Evaluation should be paired on the same held-out families and

assess both discrimination and calibration. No useful held-out advantage, serious miscalibration, or systematic failure among supposedly central members would defeat the corresponding declaration. Until such a study is completed, the forward projectibility claim remains prospective.

5.4 Category vs. function

CGEL's distinction between category and function sharpens the projectibility test: what should the INTERJECTION$_{SYN}$ classification add beyond the supplement function label? Supplements of any category (noun phrases [NPs], adjective phrases [AdjPs], relative clauses) share prosodic isolation and syntactic non-integration. The INTERJECTION$_{SYN}$ account declares a wider package:

- Non-referentiality. Supplements can be referential (*a university lecturer, Dr Brown, ...*); interjections characteristically can't.
- Emotive/interpersonal force. Not all supplements express stance (*Dr Brown* expresses none); interjections characteristically do (*ugh*, *yay*).
- Free coinage. Onomatopoeia supplies a productive route into some interjectional sign-types. Comics gave English *bam* and *pow*; the same route isn't characteristic of relative clauses or ordinary NP supplements.
- Non-inflection. NP supplements inflect freely (*a lecturer*, *two lecturers*); interjections don't (**ouches*).

The function label tells us about syntactic integration. The category classification proposes a wider family of inferences. The category/function distinction matters beyond the *CGEL* tradition: any framework that distinguishes what kind of thing an expression is from what job it's doing in a particular clause will face the same question. Whether the classification earns its keep requires the held-out comparison described above; the component contrasts motivate that test but don't substitute for it.

5.5 What sets the domain of projection

Categories don't all project in the same domain, and what fixes the domain should be where the relevant dependencies run. Nouns offer familiar morphosyntactic examples: *lecturer* takes a determiner, inflects for number, and fills an argument slot. The three interjection categories instead divide the proposed grammatical, semantic, and interactional target families among themselves.

The interactional literature motivates the hypothesis that INTERJECTION$_{PRAG}$ will support the richest projections. Dingemanse (2024) shows that frequent, functionally central interjections include continuers, repair initiators, and change-of-state tokens rather than only emotive exclamations. That makes sequential position, floor transfer, and recipient treatment natural targets. It doesn't show that INTERJECTION$_{PRAG}$ classification predicts held-out outcomes better than the local cues from which it was assigned.

Oh illustrates the proposed asymmetry. Its INTERJECTION$_{SYN}$ classification concerns packaging and distribution; its INTERJECTION$_{PRAG}$ classification concerns change-of-state and response uses. The same sign-type can make two projective families available. Its co-membership doesn't create a third source of inference.

The contrast with nouns and verbs concerns where the density lies, not how much structure each node carries. INTERJECTION$_{SYN}$ is thin next to those nodes, lacking the network of agreement, argument realization, derivation, and inflectional behaviour associated with major lexical categories. The

empirical question is whether its thinner distributional package still generalizes usefully to withheld frames.

Why might interactional targets be richest? Because routine entrenchment, if it operates as proposed in Section 4.5, connects short, standalone forms with recurrent sequential jobs. Prosodic–syntactic coupling instead motivates grammatical packaging targets. Regional or indexical conditioning may explain uneven inventory persistence but remains a maintenance hypothesis. These differently graded links motivate projections in different domains; they don't establish their accuracy.

This connects to the content/procedural distinction. The central meaning hypothesis for many interjectional sign-types is procedural and use-conditional rather than denotational (Traugott & Dasher, 2002). The INTERJECTION$_{SEM}$ category isolates that meaning contribution and makes immediacy and repeatability explicit targets. The same general relation between causal domain and projective domain appears elsewhere. ANIMACY projects grammatically (argument realization, reference tracking); COLOUR projects ecologically (edibility, toxicity) but usually doesn't project grammatically in the way animacy can (cf. Seifart, 2010).

The relative richness of the three projective families is a result for future comparison, not a premise. Nor does INTERJECTION$_{SYN}$'s grammatical standing depend on winning that comparison: its independent distributional motivation can survive failure of a proposed cross-domain edge. What would fail is the stronger claim that this classification earns additional standing through forward projectibility.

6 Discussion

6.1 Empirical probes and boundary conditions

Section 5 stated forward declarations whose held-out success remains prospective. The existing corpus and historical probes bear on different claims: recruitment, residue, country-level frequency variation, inventory turnover, and failure. They don't jointly constitute a direct test of projectibility or corrective control.

The strongest probe is pathway residue. If interjections are recruited by different causal pathways, residual properties should line up with recruitment history. The defeat condition is random distribution: retained complements, euphemistic deformation, atypical phonology, and regional skew should show no relation to source pathways. The evidence is uneven. Gehweiler's *gee* history documents an invocative route and its euphemistic phonological residue. Complement-taking *damn* is compatible with a verbal or optative source, but the sequence is reconstructive. The preregistered derivative study adds a narrower exit result: selected *wow*, *boo*, *ooh*, and *shoo* derivatives retain source-interjection semantics in their included VP-head uses. Together these findings support path dependence more strongly than they establish a category-wide causal network.

A second probe asks whether country contributes variation to item frequencies. It's weaker than testing whether conditioning decomposes a category boundary, which would require credible item-type-by-country comparisons. GloWbE frequency data (Davies & Fuchs, 2015) provide the narrower probe: for the 10 most frequent interjection-tagged items plus regional and boundary items, frequency per million was extracted by country (20 countries), and the coefficient of variation (CV; standard deviation divided by the mean) was computed across countries.

The descriptive counts show marked geographic concentration. Regional interjections (*lah*, *yaar*, *haba*) are highly concentrated: *lah* occurs at 9.35 per million in Malaysia and 6.72 in Singapore but

near zero elsewhere; *yaar* at 2.88 in Pakistan and 2.33 in India but near zero elsewhere; *haba* has a raw-string upper bound of 8.78 in Nigeria and is near zero elsewhere. Unuabonah and Daniel (2020)'s manual exclusions reduce the Nigerian count from 373 raw hits to 248 relevant tokens, approximately 5.84 per million.

Core items also vary: *ah* has the highest CV among core items (0.71), with Singapore (60.42) and Jamaica (53.16) at eight times the rate of Kenya (7.62). Even broadly distributed items like *yeah* (CV 0.55) show a 6:1 ratio between the United States (103.78) and Tanzania (16.85). Only *yes* and *no* approach uniformity (CVs of 0.21–0.22). Boundary items vary too: the fillers *um* (CV 0.61) and *uh* (CV 0.76) and the routine formula *bye* (CV 0.61) all show geographic variation.

These coefficients are estimated from twenty countries, so they carry more uncertainty than a bare figure suggests. Bootstrapping over the twenty sections identifies which comparisons survive it. The uniformity of *yes* and *no* is the firm result: their intervals sit below eleven of the thirteen other core items, overlapping only *hello* and *ha*. The ordering within the rest of the core band isn't a result at all. No two of those items have separable intervals, so *uh* at 0.76 isn't distinguishable from *yeah* at 0.55, and the figures should be read as a single undifferentiated band rather than a ranking. For the regional items the coefficient is the wrong summary altogether: with a form absent from eighteen of twenty sections the mean sits near zero and the coefficient becomes unstable, so *lah*'s interval runs from 0.89 to 4.27. Their concentration is better read off the counts than off a dispersion statistic.

Country contributes additional variation to overall item frequencies in this sample. The data doesn't test whether country effects differ systematically by item type or whether category boundaries decompose under conditioning.

A Bayesian multilevel model (negative binomial with crossed random effects for item and country) estimates positive country-level variation: the country random-effect standard deviation is 0.44 (95% credible interval [CrI]: 0.28–0.65). The estimated predictive gain over the item-only model is positive but imprecise: 18 expected log predictive density (ELPD) units with a standard error (SE) of 13 on exact leave-one-out cross-validation.

The descriptive counts and CVs are the primary evidence; the model suggests additional overall country variation under a crossed random-effects structure. Full model specification and results are in the supplementary materials.[7] Several methodological limitations remain. GloWbE's part-of-speech tagger, trained on inner-circle English, fails to tag regional interjections (*yaar* and *haba* receive no interjection tag at all), which means the tagged frequencies underestimate regional items. The *ha* row needs a criterion rather than a caveat, and the criterion has to be settled before the figure rather than after it. Iterated *HA! HA!* is a rendering of laughter, closer to eye dialect than to a lexical item; *ha* used for sarcasm or scepticism is an interjectional use. The line is harder to draw in a written corpus than a spoken one, since nothing was vocalized for the spelling to represent, and iterated *haha* in digital writing has arguably conventionalized into a form in its own right. Nothing here rests on a claim that imitative forms fall outside the category, which would contradict the recruitment route described above.

Hand-coding the tagged concordance for all twenty sections shows how much turns on that one decision, and the figures reported above count iterated laughter rather than excluding it. Laughter is 51.5% of tagged *ha* in the United States and 63.0% in Great Britain, so the two readings are far apart: across six defensible codings of the laughter and residual-token questions, *ha*'s coefficient of variation

[7] Available at https://github.com/BrettRey/interjections-hpc.

runs from 0.40 to 0.83. A single figure would report one path through that set rather than the set (cf. Gelman & Loken, 2014).

Two results should be kept apart, because one survives the spread and the other doesn't. The dispersion statistic doesn't: it isn't identified without the criterion decision, and the 0.41 above corresponds to counting laughter. The ranking does. Coding also shows what proportion of tagged tokens are interjections at all, which runs from 34.5% in Bangladesh to 84.0% in Great Britain, with HECTARE the contaminant in East Africa and South Asia and personal names in Singapore, where only 6.1% is HECTARE. Correcting each section by its own rate moves Kenya from fifth place among the twenty to fifteenth, and it does so under every one of the six codings. Pooling the twenty rates to allow for their unequal precision, which runs from about 8,100 tagged tokens for the United States down to 420 for Ghana, moves no section by more than 0.6 percentage points, because the variance between sections is much larger than the sampling variance within them.

Country differences in GloWbE may also reflect corpus composition, genre mix, internet ecology, and orthographic convention as much as dialect or community structure. A spelling-variant check bears the last of these out, in an unexpected direction. *Ya* and *yeh* look like variants of *yeah* that a single-string query would miss, but both concentrate where they're homographs of substrate words: *ya* peaks in Tanzania (19.5 per million) and Malaysia (13.9), *yeh* in Singapore (5.6), Pakistan (4.9), and India (4.9), and no *yeh* token in the corpus carries the interjection tag at all. Folding both strings into *yeah* would drop its dispersion from a CV of 0.55 to 0.46, but only by importing non-English tokens unevenly across varieties, so the figures below keep the strings apart. Lengthened spellings point the same way and put a number on it. Summing *umm*, *ummm*, and *ummmm* into *um* moves its dispersion from a CV of 0.61 to 0.60 once Pakistan is set aside, so lengthening is negligible in general; including Pakistan it rises to 0.72, because Pakistani English returns nearly eight times more *umm* than *um* where the median variety returns a third as much. A twenty-token sample settles what that signature is: nineteen are the Arabic teknonym *Umm*, as in *Umm Salama*, drawn from hadith and biographical material, and the twentieth mentions the filler rather than using it. Raw strings are weakest, then, exactly where a variety's substrate supplies a homograph, which is a sharper and more testable caveat than orthographic variation at large, and one that bears on the varieties this section compares. The regional items carry the most exposure, since they're substrate-derived to begin with. Hand-coding sixty-nine *yaar* tokens puts the English-matrix rate at roughly 60% in Pakistan and 65% in India, the remainder being Tamil *yaar* "who", proper names, and tokens sitting inside Roman-script Hindi or Urdu. The rates are estimated from twenty-token samples and are too coarse to correct with, but they're large enough that raw-string figures for these items should be read as upper bounds. Raw strings are imperfect proxies too: *yes*, *no*, *ha*, *ah*, *oh*, *well*, *right*, and *like* are not automatically interjection tokens. Some are clausal, quoted, discourse-marker uses, literary representations, or noise. The methodological payoff is narrower: the validity of tag-restricted and raw string measures varies by country, so pooled counts can hide measurement problems as well as linguistic variation.

A third, weaker probe concerns hub-property asymmetry. One property may be more central to network order than others, with prosodic isolation and non-referentiality as the main candidates. The defeat condition is equal contribution by all properties. The present evidence leaves this prediction open.

Onomatopoeic interjections like *pop* can be syntactically integrated into clause structure; when

prosodic isolation is neutralized, as when *pop* in *pop went the cork* is an adverb rather than a standalone interjection (Meinard, 2015, p. 152), reclassification is immediate. Restoring partial referentiality, by contrast, allows the form to hover between noun and interjection uses (as with *God*). This suggests prosodic isolation may be the hub, but the evidence is indirect.

The probes differ in evidential weight. Pathway residue is the main explanatory payoff, though only the invocative route is documented in detail. The GloWbE analysis establishes additional overall country variation rather than boundary decomposition, inventory counts establish turnover rather than stability, and hub-property asymmetry is an open conjecture. Prototype theory describes gradience but doesn't by itself tie independently warranted histories to source-specific residues.

At the framework level, if a feature-based account (e.g., [+peripheral, -argument-taking, +expressive]) generates the same predictions without causal structure, the network account would be a relabelling. The causal-network payoffs are pathway residues and the field-relative decomposition; if both fail under further testing, the account collapses to a descriptive property list.

6.2 Boundary disputes as causal-link overlap

The boundary disputes that embarrassed classical category theory (interjections vs. nouns, verbs, adverbs, fillers, routine formulae) become intelligible once the three field-relative cuts are separated. Each dispute may reflect partial property overlap and differences in what a field needs its classification to distinguish. The disputes motivate that analysis; they aren't independent evidence for every proposed causal link.

Routine formulae (*bye*, *hello*, *sorry*) share prosodic isolation and pragmatic function with interjections but add addressee-directedness and social predictability (Ameka, 1992, pp. 109–110). For grammar, those shared properties place them near interjections. For speech-act analysis, the additional properties matter more, because they predict conventional response slots, addressees, and felicity conditions. They're structured by additional processes (social convention, politeness norms) that prototypical interjections don't require.

Fillers (*um*, *uh*) share prosodic distinctiveness and syntactic non-integration but differ in emphasis, pause distribution, and citation-introduction ability (O'Connell & Kowal, 2005). They occupy a neighbouring region of the property space, structured by partially overlapping links (processing-pressure management rather than emotive expression).

The Ameka-Wilkins debate can be reframed in these terms. Ameka's separation preserves distinctions relevant to social routines and speech-act felicity (Ameka, 1992, p. 109). Wilkins's inclusion preserves pragmatic and semantic overlap with interjections (Wilkins, 1992, p. 142). The scope of each classification is indexed to its explanatory purpose; its adequacy still depends on the distinctions and projections the evidence supports.

7 Conclusion

The singular label interjection conceals three legitimate but non-identical classifications. $\text{interjection}_{\text{syn}}$ is principally grammar-facing, $\text{interjection}_{\text{sem}}$ semantics-facing, and $\text{interjection}_{\text{prag}}$ interaction-facing. Each cuts through a mixed network rather than containing properties from only one domain. Canonical sign-types belong to all three, but their overlap isn't a fourth category and contributes no independent projective force.

The strongest empirical result is path dependence. The history of *gee* documents an invocative route and euphemistic residue. Complement-taking *damn* is compatible with a verbal or optative

source, though the sequence remains reconstructive. Selected verbal derivatives retain source-interjection semantics in included VP-head uses. The preregistered *fie* study fails to support ordered erosion. GloWbE establishes additional country-level variation in overall item frequencies, and COHA illustrates inventory turnover; neither result establishes category-boundary decomposition or stable projectibility.

Those findings support parts of the proposed causal order and motivate the remaining links as testable hypotheses. Prosodic–syntactic coupling and routine entrenchment are stabilizer candidates; regional or indexical conditioning is a maintenance hypothesis. The evidence doesn't establish the full network, category-level stability, or corrective control.

The three-category architecture makes the forward question precise. Each classification supplies prospective declarations with independent targets: unobserved grammatical environments, semantic immediacy and repeatability, recipient uptake, next-turn treatment, and floor transfer. A held-out comparison against category-free baselines is still needed. Failure would demote the corresponding projective claim; it wouldn't by itself abolish the other categories.

The general lesson is methodological. When a traditional term spans grammar, meaning, and interaction, the choice isn't between one privileged syntactic category and an undifferentiated functional profile. Several inquiry-relative categories may be real, overlap systematically, and earn different standing. The work is to identify their bearers, keep their projections non-circular, and match each causal claim to the evidence that supports it.

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A Supplementary materials: GloWbE conditioning analysis

A.1 Data

Frequency counts for 18 items were extracted from the GloWbE corpus (Davies & Fuchs, 2015): 1.9 billion words across 20 countries. Items include the 10 most frequent interjection-tagged forms (*yes*, *no*, *oh*, *yeah*, *hi*, *hey*, *wow*, *hello*, *ah*, *ha*), three regional interjections (*lah*, *yaar*, *haba*), the semi-regional *eh*, the boundary items *cheers* and *bye*, and the fillers *um* and *uh*.

Figure 7 displays frequency per million words by country for each item. Core-item counts are tag-restricted; regional-item counts are raw strings because the tagger doesn't recognize them.

These counts are a model-dependent measurement rather than a transparent one: a tagger stands between the text and the number, so the figures carry the tagger's error as well as the corpus's composition. A model-dependent outcome has to be evaluated by variety and not only on average, since an error rate that holds overall but shifts across the varieties being compared is the measurement analogue of a trend that isn't parallel (Reynolds, 2026a). The audit reported below is that evaluation.

One limit on it should be stated plainly. Hand-coding a tagged concordance measures precision, the share of tagged tokens that are genuine, and says nothing about recall, the share of genuine tokens the tagger found. *Haba* shows recall can be zero: 373 Nigerian tokens, none tagged. So the corrected figures below are corrected for false positives only, and a section whose genuine tokens the tagger largely missed would look clean on this check while still being wrong.

Two features of the corpus constrain what country conditioning can mean here. Country is a property of the page rather than of a speaker, assigned through Google's region function: for a site with a top-level country domain the assignment is direct, but for a `.com` or `.org` site Davies and Fuchs (2015, p. 4) report that Google "relies on several heuristics, including 1) the IP address for the web server, 2) who links to that website, and 3) who visits the website", and they raise the question themselves of "how well Google has correctly identified web sites by country". The page list was also seeded by running "hundreds of very high frequency 3-grams (three word strings)" from the Corpus of Contemporary American English against Google (Davies & Fuchs, 2015, p. 4), so the sampling frame is anchored to American frequency data.

The counts aren't all of one kind either, and the reason is worth stating because it constrains what the comparison can show. English-Corpora.org tags automatically, and the tagger handles regionalisms poorly. Counts for the core items are restricted to the interjection tag, but restriction isn't the same as exclusion. Hand-coding twenty tagged *ha* tokens from Kenya and twenty from Tanzania leaves six and eight respectively as interjections; the rest are almost all the abbreviation for HECTARE, in agricultural and development prose. On that basis *ha* in Kenya falls from 24.9 tokens per million to roughly 7.5, and from fifth place among the twenty countries to nineteenth. So the tag narrows a count without cleaning it, and it fails worst where a homograph belongs to a text type a region produces more of.

Counts for *haba* and *yaar* can't be tag-restricted at all, because neither string is ever assigned that tag: all 373 Nigerian tokens of *haba* are tagged as common or proper nouns, 65% of them on a reading that includes the proper noun. The regional figures are raw strings where the core figures are tag-restricted, which inflates the regional items relative to the items they're set beside. False positives in the tagged core counts partially mask the mismatch. Its scale is indicated by Unuabonah and Daniel (2020), who work on *haba* in the same Nigerian component and report 248 tokens against the 373 a raw query returns. They reach the smaller figure by inspecting each hit and excluding Nigerian

Pidgin, indigenous-language, and duplicated material (Unuabonah & Daniel, 2020, p. 69). Two of those three exclusions are invisible to a frequency query, and the third, duplication, is a known property of the corpus. That's a limit imposed by the tagger rather than a choice: a form has to be known to the tagger before it can be tagged, and a borrowed interjection in a postcolonial variety is exactly what an automatic tagger trained elsewhere will miss.

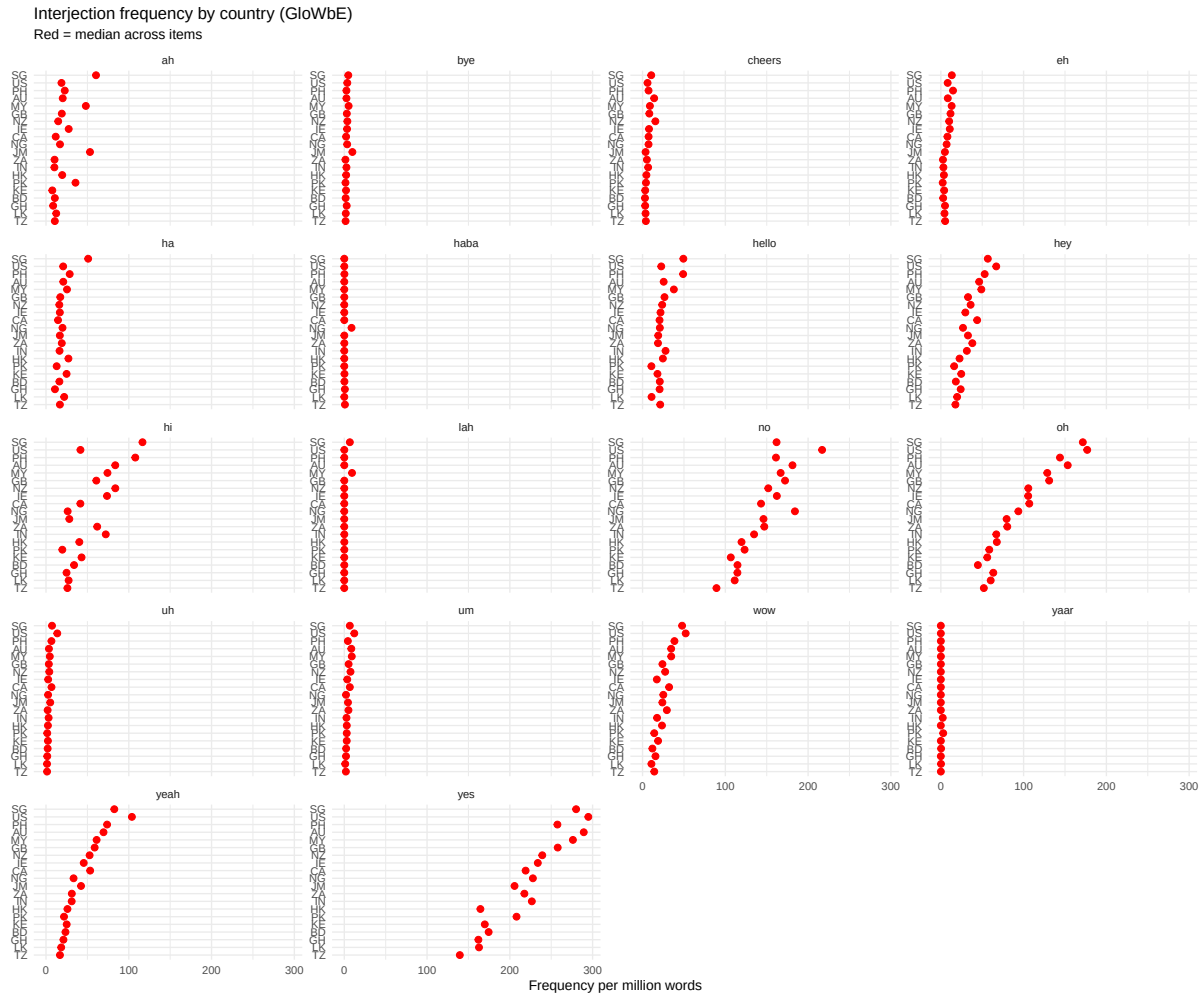


Figure 7: Interjection frequency per million words across 20 GloWbE countries. Red points indicate the median across items within each country. Regional items (*lah*, *yaar*, *haba*) cluster in one or two countries. Core-item frequencies also vary across sections, while only *yes* and *no* approach uniformity. Tagging and raw-string validity differ by item and country, as discussed in the text.

A.2 Model specification

Four Bayesian negative binomial models were fitted using `brms` 2.23 in R, with 4 chains of 2,000 iterations (1,000 warmup) each. The confirmatory models (m0–m2) use exact reloo for leave-one-out cross-validation; the exploratory model (m3) is assessed by Pareto-smoothed importance sampling (PSIS) diagnostics only.

`m0` (unconditional):

```
count ~ 1 + offset(log(words_m)) + (1 | item)
```

`m1` (conditioned):

```
count ~ 1 + offset(log(words_m)) +
(1 | item) + (1 | country)
```

`m2` (item type):

```
count ~ 1 + item_type + offset(log(words_m)) +
(1 | item) + (1 | country)
```

`m3` (exploratory):

```
count ~ 1 + item_type + offset(log(words_m)) +
(1 | item) + (1 + item_type || country)
```

Model m3 estimates country-level varying slopes by item type. Because several item-type cells are sparse (semi-regional: 1 item, boundary: 2, filler: 2, regional: 3), these slopes are thinly supported and m3 is treated as exploratory.

A.3 Results

Exact leave-one-out cross-validation (confirmatory models):

Model	ELPD diff	SE diff
m1 (conditioned)	0.0	0.0
m2 (item type)	−9.3	7.9
m0 (unconditional)	−17.9	13.3

The conditioned model (m1) has the highest ELPD point estimate, but its estimated advantage over m0 is imprecise. Adding item-type fixed effects (m2) doesn't improve the point estimate over m1, while the item random effect captures substantial between-item variation. The exploratory model (m3) is excluded from formal ranking because 9 observations retain Pareto $k > 0.7$ after moment matching. The corresponding PSIS leave-one-out estimates are unstable.

Random effects from the conditioned model (m1):

- Country standard deviation (SD) = 0.44 (95% CrI: 0.28–0.65)
- Item SD = 2.07 (95% CrI: 1.49–2.96)
- Ratio: 4.7:1 (item identity varies more, while country contributes additional overall variation)

The evidence is directionally consistent: conditioning on country has a positive estimated predictive difference (ΔELPD = 17.9, SE 13.3), and the country random-effect SD is positive across the interval. The SE is large relative to the point estimate in this sample (18 items × 20 countries = 360 observations). The descriptive CV results in the main text provide the primary evidence for the conditioning probe; the model suggests the pattern survives the crossed random-effects structure.

The hand audit of *ha* in Section 6 shows that its counts carry a country-varying validity rate. To propagate that error, each country's rate thins *ha* by a draw from its BETA posterior; the other seventeen items stay unchanged, and both models are refitted across draws. The correction is small. The country random-effect SD shifts from 0.42 to 0.44 with iterated laughter counted and 0.45 with it excluded; the model comparison shifts from 28.8 to 26.9 and 28.5. These approximate leave-one-out values permit only an internal contrast. Since concordance coding estimates precision, the correction removes false positives, not missed tokens.

B Supplementary materials: semantic residue in verbal derivatives (COHA)

B.1 Data and extraction

The morphology-gain study was pre-registered before extraction and coding (`f1a2193`, 27 May 2026). The corpus was COHA (1820–2019). The fixed source families were *wow*, *boo*, *ooh*, and *shoo*. The fixed target derivatives were *wowed*, *wowing*, *wows*; *booed*, *booing*, *boos*; *oohed*, *oohing*, *oohs*; and *shooed*, *shooing*, *shoos*. Bare source forms were excluded.

Extraction was completed on 27 May 2026. The token-level analysis used parsed key-word-in-context (KWIC) exports. Chart requests for *oohed* and *oohing* returned server errors, but KWIC exports loaded and were used for coding. Minor chart/KWIC row-count discrepancies are documented in the extraction summary.

The go/no-go threshold required at least 20 analyzable derivative tokens and at least two decades with three or more analyzable derivative tokens. All four families met the threshold.

Family	Analyzable tokens	Decades with 3+	Status
boo	364	10	GO
ooh	52	6	GO
shoo	410	13	GO
wow	143	9	GO

B.2 Coding

Each analyzable token was coded for two confirmatory properties. First, verbal syntax: whether the derivative heads a VP. Second, semantic residue: whether the derivative retains the source interjection's meaning profile: impressed or evaluative response for *wow*; disapproval or contempt for *boo*; appreciative or surprised response for *ooh*; and directive away-sending for *shoo*.

Two secondary descriptive properties were also coded. Interactional residue tracks whether the verbal event retains the source interjection's social-interactional profile as a recognizable act of response, uptake, display, or direction. Targeted entity tracks whether a target, object, addressee, or stimulus is recoverable. These secondary codes weren't used in the confirmatory test.

B.3 Confirmatory result

For each source family, the confirmatory window was the earliest decade with at least three analyzable derivative tokens, plus following decades until the window contained at least ten tokens. The success criterion was `verbal_syntax = 1` and `semantic_residue = 1`. The pre-registered support criterion required the lower bound of the Jeffreys 95% interval to exceed 0.5.

Family	Tokens	Window	n	Successes	Jeffreys 95% CI
boo	364	1920s	17	17	[0.865, 1.000]
ooh	52	1960s–1980s	10	10	[0.783, 1.000]
shoo	410	1890s–1900s	16	16	[0.857, 1.000]
wow	143	1930s–1940s	11	11	[0.800, 1.000]

All four families meet the preregistered support criterion. The inference is conditional on the design: source families were selected through overt derivatives, and the success criterion required both VP-head syntax and semantic residue. The result concerns semantic persistence in these selected derivative VP-head uses, not a general tendency for interjections to gain morphology or syntax. The secondary diachronic model wasn't estimated because semantic residue had no variation among included VP-head tokens.

B.4 Reliability and files

A blind 10% recode was completed ($n = 97$, seed 2026). The confirmatory codes were fully reliable: inclusion, VP-head status, and semantic residue each had agreement = 1.00 and Cohen's $\kappa = 1.00$. The secondary descriptive codes were less stable.

Code	Agreement	Cohen's κ
Inclusion	1.000	1.000
VP-head status	1.000	1.000
Semantic residue	1.000	1.000
Interactional residue	0.887	0.297
Targeted entity	0.557	0.025

The analysis files are in `analysis/morphgain-confirmatory/`. The main outputs are:

- `morphgain-extraction-summary.csv`
- `morphgain-go-nogo-summary.csv`
- `morphgain-coding-summary.csv`
- `morphgain-confirmatory-results.csv`
- `morphgain-earliest-windows.csv`
- `morphgain-reliability-results.csv`

The scripts are:

- `morphgain-confirmatory-extract.mjs`
- `morphgain-confirmatory-prepare.py`
- `morphgain-confirmatory-code-primary.py`
- `morphgain-confirmatory-analyze.py`
- `morphgain-confirmatory-reliability.py`

C Supplementary materials: obsolescence analysis (*fie* in COHA)

C.1 Data and coding

616 tokens of *fie* were extracted from COHA (1820–2019), excluding 141 optical character recognition (OCR) artefacts (where *fie* is a misread of *he*, *the*, or *five*). Each token was coded for three binary node-membership diagnostics: syntactic, semantic/use-conditional, and interactional-pragmatic interjectionhood. These correspond to the $_{\text{syn}}$, $_{\text{sem}}$, and $_{\text{prag}}$ nodes used above. Coding criteria are in the pre-registration. The decade column was hidden during coding.

Intra-coder reliability (10% blind recode, $n = 62$):

Diagnostic	Agreement	Cohen's κ
INTERJECTION$_{\text{SYN}}$	87.1%	0.68
INTERJECTION$_{\text{SEM}}$	96.8%	0.90
INTERJECTION$_{\text{PRAG}}$	96.8%	0.90

C.2 Raw proportions

Figure 4 shows the proportion of tokens retaining each diagnostic by decade. All three diagnostics are near ceiling (85–100%) through the 1890s, decline sharply between 1900 and 1940, and fluctuate at low levels thereafter. The late rebounds (1980s, 2000s) reflect historical fiction and Shakespeare adaptations in which characters use *fie* as a live interjection.

C.3 Confirmatory model

Bayesian logistic decline models (per the pre-registration) estimate the 50% effective date (ED50; τ: the year at which each diagnostic's probability crosses 0.5):

Diagnostic	τ (median)	95% CrI
INTERJECTION$_{\text{SYN}}$	1947	1937–1958
INTERJECTION$_{\text{PRAG}}$	1948	1939–1959
INTERJECTION$_{\text{SEM}}$	1949	1941–1960

Pre-committed prediction: $\Pr(\tau_{\text{int}} < \tau_{\text{sem}} < \tau_{\text{syn}}) = 0.13$ (threshold for "inconsistent": < 0.25). The predicted sequential erosion wasn't observed. The three marginal decline distributions aren't distinguished by this analysis. See Figure 8.

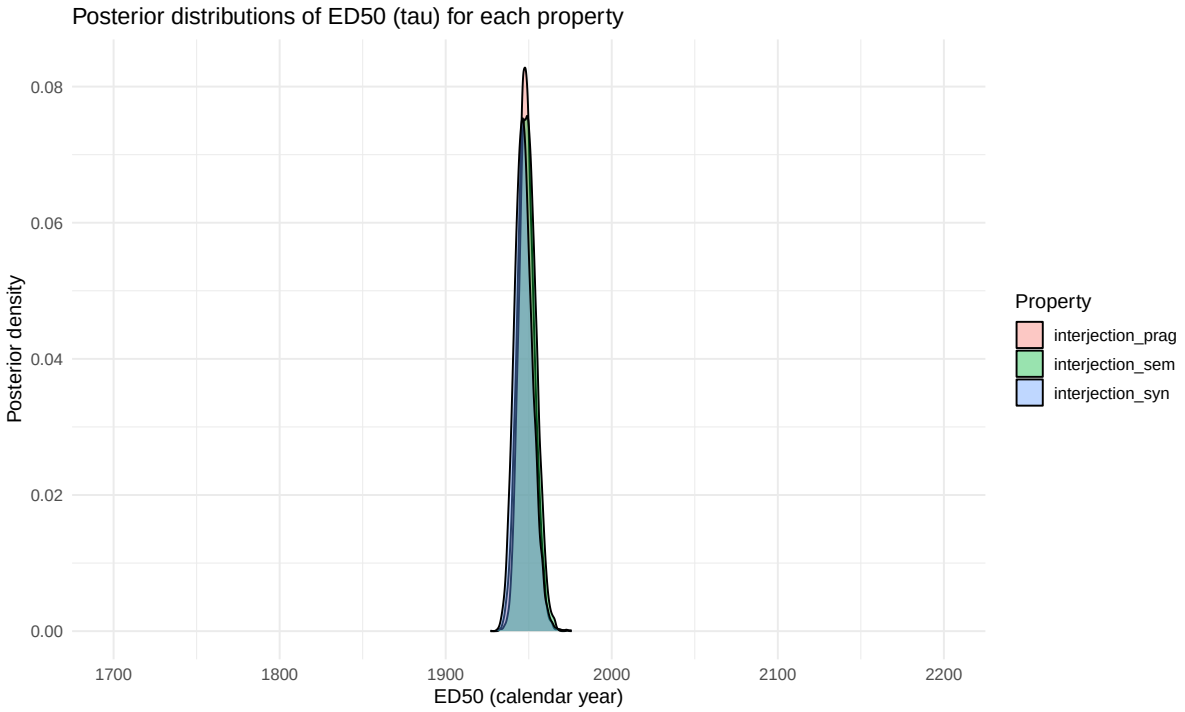


Figure 8: Posterior distributions of ED50 (τ) for each diagnostic. The pre-registered ordered-erosion prediction received low posterior support, and the three marginal ED50 distributions overlap extensively. Their overlap shows failure to distinguish the dates, not equality of the dates.